

Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 1 / 77
Rev.: 00

CE EMC TEST REPORT

For

Micro SD card

Model No.: DTTF1TU3A2V30AP

Added Model(s):

DTTF1.5TU3A2V30AP, DTTF1TU3A2V30A, DTTF1.5TU3A2V30A,
DTTF512GU3A2V30HA, DTTF512GU3A2V30AP, DTTF512GU3A2V30A,
DTTF256GU3A2V30HA, DTTF256GU3A2V30AP, DTTF256GU3A2V30A,
DTTF128GU3A2V30HA, DTTF128GU3A2V30AP, DTTF128GU3A2V30A

Issued to:

DATOTEK INTERNATIONAL Co., Ltd.

No. 34, Alley 18, Lane 478, Rueiguang Rd., Neihu District, Taipei, Taiwan

Issued by:

Compliance Certification Services Inc.

Wugu EMC Laboratory

No.139, Wugong Rd., Wugu Dist., New Taipei City, Taiwan.

Issued Date: April 02, 2025

Note: This document may be altered or revised by Wugu EMC Laboratory of CCS Inc. personnel only, and shall be noted in the revision section of the document. The client should not use it to claim product endorsement by TAF, NIST or any government agencies. The test results in the report only apply to the tested sample.

Unless otherwise stated the results shown in this test report refer only to the sample(s) tested and such sample(s) are retained for 90 days only.

除非另有說明，此報告結果僅對測試之樣品負責，同時此樣品僅保留90天。本報告未經本公司書面許可，不可部份複製。

This document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at <http://www.sgs.com.tw/Terms-and-Conditions> and, for electronic format documents, subject to Terms and Conditions for Electronic Documents at <http://www.sgs.com.tw/Terms-and-Conditions>. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein. Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of client's instruction, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced, except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law.



Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 2 / 77
Rev.: 00

Revision History

Rev.		Issue Date		Revisions	Effect Page	Revised By
00		April 02, 2025		Initial Issue	ALL	Allison Chen

Note:
Rev.(00)
1. TMWK2408002963KE licensed to TMWK2503001385KE. All construction, materials, components are the same.

DRAFT

TABLE OF CONTENTS

1	TEST CERTIFICATION.....	4
2	TEST RESULT SUMMARY	5
3	EUT DESCRIPTION	6
4	TEST METHODOLOGY	7
4.1.	DECISION OF FINAL TEST MODE	7
4.2.	EUT SYSTEM OPERATION	7
5	SETUP OF EQUIPMENT UNDER TEST	8
5.1.	DESCRIPTION OF SUPPORT UNITS	8
5.2.	CONFIGURATION OF SYSTEM UNDER TEST	9
6	FACILITIES AND ACCREDITATIONS	10
6.1.	FACILITIES	10
6.2.	ACCREDITATIONS	10
6.3.	MEASUREMENT UNCERTAINTY	11
7	EMISSION TEST	12
7.1.	CONDUCTED EMISSION MEASUREMENT AT AC MAINS PORT	12
7.2.	REQUIREMENTS FOR ASYMMETRIC MODE CONDUCTED EMISSIONS	17
7.3.	RADIATED EMISSION MEASUREMENT	21
7.4.	CONDUCTED DIFFERENTIAL VOLTAGE EMISSIONS FROM CLASS B EQUIPMENT	33
7.5.	HARMONICS CURRENT MEASUREMENT	36
7.6.	VOLTAGE FLUCTUATION AND FLICKER MEASUREMENT	39
8	IMMUNITY TEST	42
8.1.	GENERAL DESCRIPTION	42
8.2.	GENERAL PERFORMANCE CRITERIA DESCRIPTION	43
8.3.	ELECTROSTATIC DISCHARGE (ESD)	44
8.4.	RADIATED, RADIO-FREQUENCY, ELECTROMAGNETIC FIELD (RS).....	48
8.5.	ELECTRICAL FAST TRANSIENT (EFT)	52
8.6.	SURGE IMMUNITY TEST	55
8.7.	CONDUCTED RADIO FREQUENCY DISTURBANCES (CS)	58
8.8.	POWER FREQUENCY MAGNETIC FIELD	62
8.9.	VOLTAGE DIP & VOLTAGE INTERRUPTIONS	65
9	PHOTOGRAPHS OF THE TEST CONFIGURATION.....	68
	APPENDIX 1 - PHOTOGRAPHS OF EUT	76

1 TEST CERTIFICATION

Product: Micro SD card

Brand: DATO

Model No.: DTF1TU3A2V30AP

Added Model(s): DTF1.5TU3A2V30AP, DTF1TU3A2V30A, DTF1.5TU3A2V30A,
DTF512GU3A2V30HA, DTF512GU3A2V30AP, DTF512GU3A2V30A,
DTF256GU3A2V30HA, DTF256GU3A2V30AP, DTF256GU3A2V30A,
DTF128GU3A2V30HA, DTF128GU3A2V30AP, DTF128GU3A2V30A

Applicant: DATOTEK INTERNATIONAL Co., Ltd.

No. 34, Alley 18, Lane 478, Rueiguang Rd., Neihu District, Taipei, Taiwan

Manufacturer: Phison Electronics Corp.

No. 1, Qun Yi Rd., Jhunan, Miaoli 350, Taiwan

Tested: August 23 ~ September 03, 2024

Test Voltage: AC 230V, 50Hz

Applicable Standards: EN 55032: 2015+A11: 2020
EN IEC 61000-3-2: 2019
EN 61000-3-3: 2013+A1: 2019

EN 55035: 2017+A11: 2020
IEC 61000-4-2: 2008
IEC 61000-4-3: 2020
IEC 61000-4-4: 2012
IEC 61000-4-5: 2014+A1: 2017
IEC 61000-4-6: 2013+COR1: 2015
IEC 61000-4-8: 2009
IEC 61000-4-11: 2020+COR1: 2020

Deviation from Applicable Standard

None

Statements of Conformity

Determination of compliance is based on the results of the compliance measurement, not taking into account measurement instrumentation uncertainty.

This test report can be used for CE and UKCA marking application which is based on equivalent requirements between UK and EU. It is appropriate using designated standards to provide presumption of conformity with GB law.

The above equipment has been tested by Wugu EMC Laboratory of CCS Inc. and found compliance with the requirements set forth in the technical standards mentioned above. The results of testing in this report apply only to the product/system, which was tested. Other similar equipment will not necessarily produce the same results due to production tolerance and measurement uncertainties.

Approved by:



Hex Chiang
Manager

2 TEST RESULT SUMMARY

EMISSION			
Standard	Item	Result	Remarks
EN 55032: 2015+A11: 2020	Conducted (Power Port)	PASS	Meet Class B limit
	Asymmetric mode conducted emissions	N/A	Not applicable, the EUT doesn't have LAN Port or Modem port.
	Radiated	PASS	Meet Class B limit
	Radiated emissions from FM receivers	N/A	Not applicable, the EUT doesn't have FM receiver's functionality.
	Conducted differential voltage emissions from Class B equipment	N/A	Not applicable, the EUT is not (TV broadcast receiver / RF modulator output ports / FM broadcast receiver)
EN IEC 61000-3-2: 2019	Harmonic current emissions	PASS	Meet Class A limit
EN 61000-3-3: 2013+A1: 2019	Voltage fluctuations & flicker	PASS	Meets the requirements

IMMUNITY 【EN 55035: 2017+A11: 2020】			
Standard	Item	Result	Remarks
IEC 61000-4-2: 2008	ESD	PASS	Meets the requirements of Performance Criterion B
IEC 61000-4-3: 2020	RS	PASS	Meets the requirements of Performance Criterion A
IEC 61000-4-4: 2012	EFT	PASS	Meets the requirements of Performance Criterion B
IEC 61000-4-5: 2014+A1: 2017	Surge	PASS	Meets the requirements of Performance Criterion B
IEC 61000-4-6: 2013+COR1: 2015	CS	PASS	Meets the requirements of Performance Criterion A
IEC 61000-4-8: 2009	PFMF	PASS	Meets the requirements of Performance Criterion A
IEC 61000-4-11: 2020+COR1: 2020	Voltage dips & voltage variations	PASS	Meets the requirements of Voltage Dips: 1) >95% reduction Performance Criterion B 2) 30% reduction Performance Criterion C Voltage Interruptions: 1) >95% reduction Performance Criterion C

Note: 1. The statements of test result on the above are decided by the request of test standard only; the measurement uncertainties are not factored into this compliance determination.
2. The information of measurement uncertainty is available upon the customer's request.

3 EUT DESCRIPTION

Product	Micro SD card
Brand Name	DATO
Model No.	DTTF1TU3A2V30AP
Added Model(s)	DTTF1.5TU3A2V30AP, DTTF1TU3A2V30A, DTTF1.5TU3A2V30A, DTTF512GU3A2V30HA, DTTF512GU3A2V30AP, DTTF512GU3A2V30A, DTTF256GU3A2V30HA, DTTF256GU3A2V30AP, DTTF256GU3A2V30A, DTTF128GU3A2V30HA, DTTF128GU3A2V30AP, DTTF128GU3A2V30A
Applicant	DATOTEK INTERNATIONAL Co., Ltd.
Housing material	Plastic
Received Date	August 22, 2024
EUT Power Rating	DC 3.3V
AC Power Cord Type	N/A
DC Power Cord Type	N/A
Controller	PS8229
PCB	PG-HS-008-5601B-FN;PG-HS-008-XXXX-XX (X=0~9,A~Z)

Note:

1. Difference of the model numbers (list on this report) is just for different capacity as below:

Model	Capacity
DTTF1TU3A2V30AP	1TB
DTTF1.5TU3A2V30AP	1TB
DTTF1TU3A2V30A	1TB
DTTF1.5TU3A2V30A	1TB
DTTF512GU3A2V30HA	512GB
DTTF512GU3A2V30AP	512GB
DTTF512GU3A2V30A	512GB
DTTF256GU3A2V30HA	256GB
DTTF256GU3A2V30AP	256GB
DTTF256GU3A2V30A	256GB
DTTF128GU3A2V30HA	128GB
DTTF128GU3A2V30AP	128GB
DTTF128GU3A2V30A	128GB

2. Disclaimer:

Variant information between/among model numbers / trademarks are provided by the applicant, test results of this report are applicable to the sample EUT received of main test model name.

I/O Port

I/O PORT TYPES	Q'TY	TESTED WITH
N/A		

4 TEST METHODOLOGY

4.1. DECISION OF FINAL TEST MODE

The EUT was tested together with the above additional components, and a configuration, which produced the worst emission levels, was selected and recorded in this report.

1. The following test mode(s) was(were) scanned during the preliminary test:

Pre-Test Mode
Mode 1: EUT (DTTF1TU3A2V30AP), Normal Operation Mode

2. After the preliminary scan, the following test mode(s) was(were) found to produce the highest emission level.

Final Test Mode		
Emission	Conducted Emission	Mode 1
	Radiated Emission	Mode 1
PH & PF		Mode 1
Immunity		Mode 1

Then, the above highest emission mode of the configuration of the EUT and cable was chosen for all final test items.

4.2. EUT SYSTEM OPERATION

1. Setup the EUT as shown on 5.2.
2. Turn on the power of all equipment.
3. The Notebook PC will reads/writes disk through “WINEMC” test software.
4. The Notebook PC displays television color bar signal through “Windows Media Player” (File name ITU-R BT 471-1).”.
5. Setup the condition for test mode and begin the test.

Note: Test program is self-repeating throughout the test.

5 SETUP OF EQUIPMENT UNDER TEST

5.1. DESCRIPTION OF SUPPORT UNITS

The EUT has been tested as an independent unit together with other necessary accessories or support units. The following support units or accessories were used to form a representative test configuration during the tests.

Peripherals Devices:

No.	Equipment	Trade Name	Model No.	Serial No.
1	Notebook PC	Lenovo	TP00103G	PF-24WF25
2	Mouse	HP	m150	B1M1502004000394
3	Printer	EPSON	Stylus-C63	FAPY150360

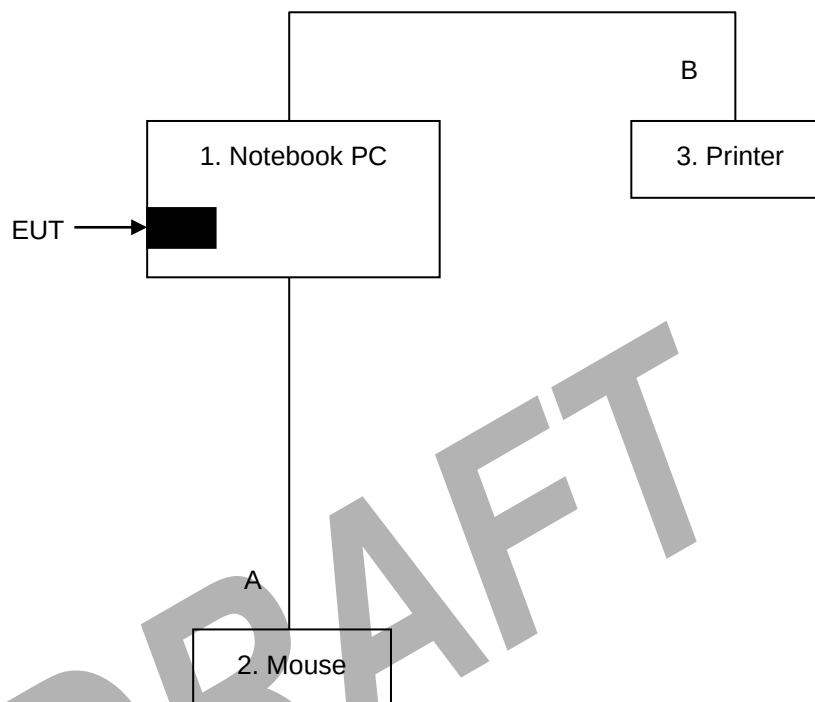
No.	Cable Name	Unit	Shielded	Length	With Core
(A)	USB Mouse Cable	1	☒ Shielded, ☐ Non	1.8 m	☐ With Core×____, ☒ Non
(B)	USB Printer Cable	1	☒ Shielded, ☐ Non	1.8 m	☐ With Core×____, ☒ Non

Note:

- 1) All the equipment/cables were placed in the worst-case configuration to maximize the emission during the test.
- 2) Grounding was established in accordance with the manufacturer's requirements and conditions for the intended use.

5.2. CONFIGURATION OF SYSTEM UNDER TEST

Mode 1:



6 FACILITIES AND ACCREDITATIONS

6.1. FACILITIES

All measurement facilities used to collect the measurement data are located at:

No.139, Wugong Rd., Wugu Dist., New Taipei City, Taiwan.

The sites are constructed in conformance with the requirements of ANSI C63.4 and CISPR 32. All receiving equipment conforms to CISPR 16-1-1, CISPR 16-1-2, CISPR 16-1-4, CISPR 16-2-1, CISPR 16-2-3, CISPR 16-4-2.

6.2. ACCREDITATIONS

Our laboratories are accredited and approved by the following accreditation body according to ISO/IEC 17025.

Taiwan	TAF 4265
---------------	----------

The measuring facility of laboratories has been authorized or registered by the following approval agencies.

Japan	VCCI Radiated emissions: 30 MHz -1000 MHz: R-14343 / Above 1GHz: G-10945 Conducted Emission B: C-13700 / T-11839
USA	FCC (10M Semi Anechoic Chamber (FCC MRA: TW1150) to perform FCC Part 15 measurements)

Copies of granted accreditation certificates are available for downloading from our web site.

6.3. MEASUREMENT UNCERTAINTY

Where relevant, the following measurement uncertainty levels have been estimated for tests performed on the EUT as specified in CISPR 16-4-2:

Measurement	Frequency	Uncertainty
Conducted Test Site No.B	0.15MHz ~ 30MHz	± 3.37 dB
Conducted Test Site No.B (Telecommunication Ports)	0.15MHz ~ 30MHz	± 3.35 dB
Radiated emissions (10M Chamber)	30MHz ~ 1GHz	± 4.41 dB
	Above 1GHz	± 4.97 dB

This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of $k=2$.

Consistent with industry standard (e.g. CISPR 32: 2015, clause 11, Measurement Uncertainty) determining compliance with the limits shall be based on the results of the compliance measurement. Consequently, the measured emissions being less than the maximum allowed emission results in this be a compliant test or passing test.

The listed uncertainties of above table are the worst-case values for the entire range of measurement. Please note that the uncertainty values are only provided for informational purposes and aren't used in determining the PASS/FAIL results.

7 EMISSION TEST

7.1. CONDUCTED EMISSION MEASUREMENT AT AC MAINS PORT

7.1.1. LIMITS

FREQUENCY (MHz)	Class A (dBuV)		Class B (dBuV)	
	Quasi-peak	Average	Quasi-peak	Average
0.15 - 0.5	79	66	66 - 56	56 - 46
0.50 - 5.0	73	60	56	46
5.0 - 30.0	73	60	60	50

Note:

1. The lower limit shall apply at the transition frequencies.
2. The limit decreases in line with the logarithm of the frequency in the range of 0.15 to 0.50MHz.
3. All emanations from a class A/B digital device or system, including any network of conductors and apparatus connected thereto, shall not exceed the level of field strengths specified above.

7.1.2. TEST INSTRUMENTS

Conducted Test Site No.B					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
Cable	EMCI	EMCRG142	190809	2024/04/21	2025/04/20
LISN	SCHWARZBECK	NSLK8128	8128-5012	2024/05/21	2025/05/20
LISN	SCHWARZBECK	NSLK8127	01022	2024/05/07	2025/05/06
EMI Test Receiver	R&S	ESR3	102915	2023/10/30	2024/10/29
Pulse Limiter	SCHWARZBECK	VTSD 9561-F	00817	2024/04/21	2025/04/20
Software	CCS-3A1-CE				

Note: The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.

7.1.3. TEST PROCEDURES (please refer to measurement standard or CCS SOP PA-031)

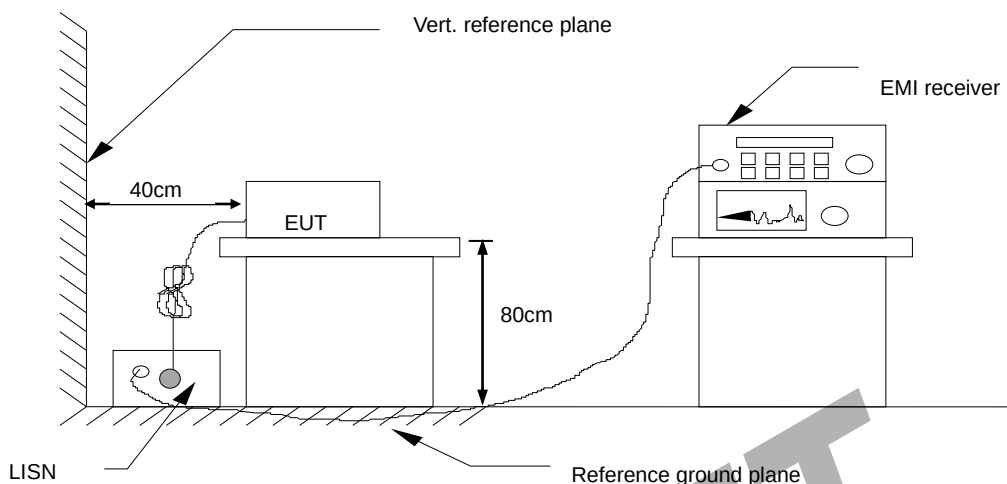
Procedure of Preliminary Test

- The EUT and Support equipment, if needed, was set up as per the test configuration to simulate typical usage per the user's manual. When the EUT is a tabletop system, a wooden table with a height of 0.8 meters is used and is placed on the ground plane as per EN 55032 (see Test Facility for the dimensions of the ground plane used). When the EUT is a floor standing equipment, it is placed on the ground plane, which has a 15 cm non-conductive covering to insulate the EUT from the ground plane.
- All I/O cables were positioned to simulate typical actual usage as per EN 55032.
- The test equipment EUT installed received AC main power, through a Line Impedance Stabilization Network (LISN), which supplied power source and was grounded to the ground plane.
- All support equipment power received from a second LISN.
- The EUT test program was started. Emissions were measured on each current carrying line of the EUT using an EMI Test Receiver connected to the LISN powering the EUT.
- The Receiver scanned from 150kHz to 30MHz for emissions in each of the test modes.
- During the above scans, the emissions were maximized by cable manipulation.
- The test mode(s) described in Item 4.1 were scanned during the preliminary test.
- After the preliminary scan, we found the test mode described in Item 4.1 producing the highest emission level.
- The EUT configuration and cable configuration of the above highest emission levels were recorded for reference of the final test.

Procedure of Final Test

- EUT and support equipment were set up on the test bench as per the configuration with highest emission level in the preliminary test.
- A scan was taken on both power lines, Line 1 and Line 2, recording at least the six highest emissions. Emission frequency and amplitude were recorded into a computer in which correction factors were used to calculate the emission level and compare reading to the applicable limit.
- The test data of the worst-case condition(s) was recorded.

7.1.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

7.1.5. DATA SAMPLE

Freq. (MHz)	Reading (dBuV)	Factor (dB)	Result (dBuV)	Limit (dBuV)	Margin (dB)	Detector (P/Q/A)	Line (L1/L2)
x.xx	42.95	0.55	43.50	56	-12.50	Q	L1

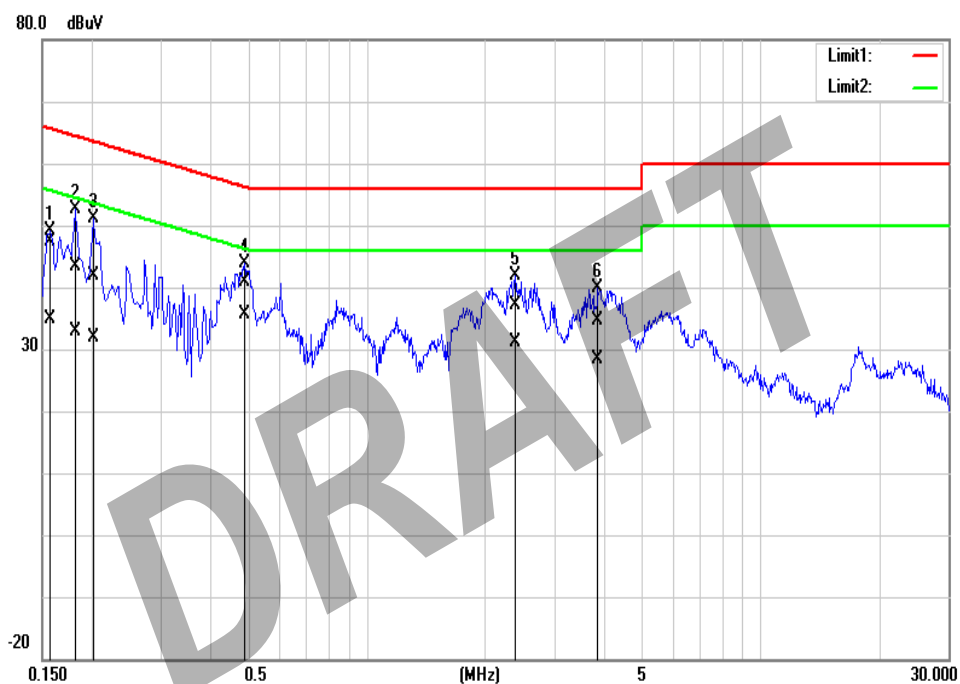
Freq.	= Emission frequency in MHz
Reading	= Uncorrected Analyzer/Receiver reading
Factor	= Insertion loss of LISN + Cable Loss + Pulse Limit
Result	= Reading + Factor
Limit	= Limit stated in standard
Margin	= Reading in reference to limit
P	= Peak Reading
Q	= Quasi-peak Reading
A	= Average Reading
L1	= Hot side
N	= Neutral side

Calculation Formula

$$\text{Margin (dB)} = \text{Result (dBuV)} - \text{Limit (dBuV)}$$

7.1.6. TEST RESULTS

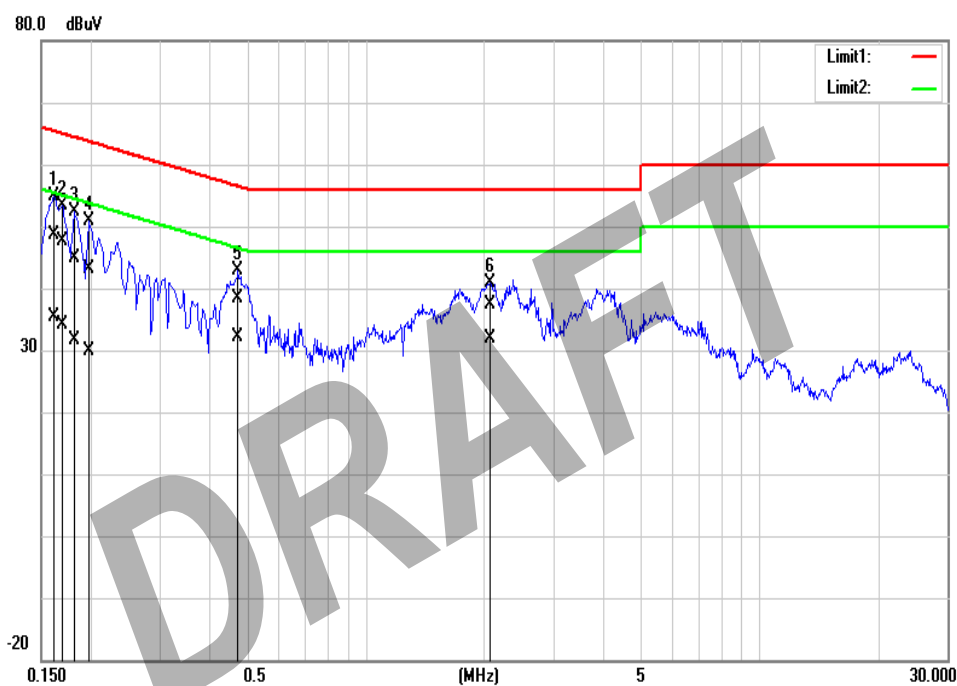
Model No.	DTTF1TU3A2V30AP	Line:	L1
Environmental Conditions	23°C, 47% RH	Tested Date	2024/08/23
Test Mode	Mode 1	Tested By	Thomas Liu
6dB Bandwidth	9 kHz		



NO.	Frequency	Quasi Peak reading	Average reading	Correction factor	Quasi Peak result	Average result	Quasi Peak limit	Average limit	Quasi Peak margin	Average margin	Remark
	(MHz)	(dBuV)	(dBuV)	(dB)	(dBuV)	(dBuV)	(dBuV)	(dBuV)	(dB)	(dB)	(Pass/Fail)
1	0.1580	37.09	24.76	10.22	47.31	34.98	65.56	55.57	-18.25	-20.59	Pass
2	0.1819	33.09	22.64	10.22	43.31	32.86	64.39	54.40	-21.08	-21.54	Pass
3	0.2020	31.54	21.68	10.22	41.76	31.90	63.52	53.53	-21.76	-21.63	Pass
4*	0.4900	30.60	25.45	10.25	40.85	35.70	56.17	46.17	-15.32	-10.47	Pass
5	2.3860	26.81	20.71	10.31	37.12	31.02	56.00	46.00	-18.88	-14.98	Pass
6	3.8660	24.16	18.03	10.35	34.51	28.38	56.00	46.00	-21.49	-17.62	Pass

REMARKS: L1 = Live Line

Model No.	DTTF1TU3A2V30AP	Line:	N
Environmental Conditions	23°C, 47% RH	Tested Date	2024/08/23
Test Mode	Mode 1	Tested By	Thomas Liu
6dB Bandwidth	9 kHz		



NO.	Frequency (MHz)	Quasi Peak reading (dBuV)	Average reading (dBuV)	Correction factor (dB)	Quasi Peak result (dBuV)	Average result (dBuV)	Quasi Peak limit (dBuV)	Average limit (dBuV)	Quasi Peak margin (dB)	Average margin (dB)	Remark (Pass/Fail)
1	0.1620	38.51	25.20	10.22	48.73	35.42	65.36	55.36	-16.63	-19.94	Pass
2	0.1700	37.51	23.84	10.22	47.73	34.06	64.96	54.96	-17.23	-20.90	Pass
3	0.1819	34.59	21.31	10.23	44.82	31.54	64.39	54.40	-19.57	-22.86	Pass
4	0.1980	32.86	19.77	10.23	43.09	30.00	63.69	53.69	-20.60	-23.69	Pass
5	0.4740	28.19	21.91	10.25	38.44	32.16	56.44	46.44	-18.00	-14.28	Pass
6*	2.0740	27.02	21.66	10.31	37.33	31.97	56.00	46.00	-18.67	-14.03	Pass

REMARKS: N = Neutral Line

7.2. REQUIREMENTS FOR ASYMMETRIC MODE CONDUCTED EMISSIONS

7.2.1. LIMITS

For Class A Equipment

FREQUENCY (MHz)	Voltage Limit (dBuV)		Current Limit (dBuA)	
	Quasi-peak	Average	Quasi-peak	Average
0.15 ~ 0.5	97 ~ 87	84 ~ 74	53 ~ 43	40 ~ 30
0.5 ~ 30.0	87	74	43	30

NOTE: The limits decrease linearly with the logarithm of the frequency in the range 0.15 MHz to 0.5 MHz

For Class B Equipment

FREQUENCY (MHz)	Voltage Limit (dBuV)		Current Limit (dBuA)	
	Quasi-peak	Average	Quasi-peak	Average
0.15 - 0.5	84 ~ 74	74 ~ 64	40 ~ 30	30 ~ 20
0.5 - 30.0	74	64	30	20

NOTE: The limits decrease linearly with the logarithm of the frequency in the range 0.15 MHz to 0.5 MHz

7.2.2. TEST INSTRUMENTS

Conducted Test Site No.B					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
N/A					

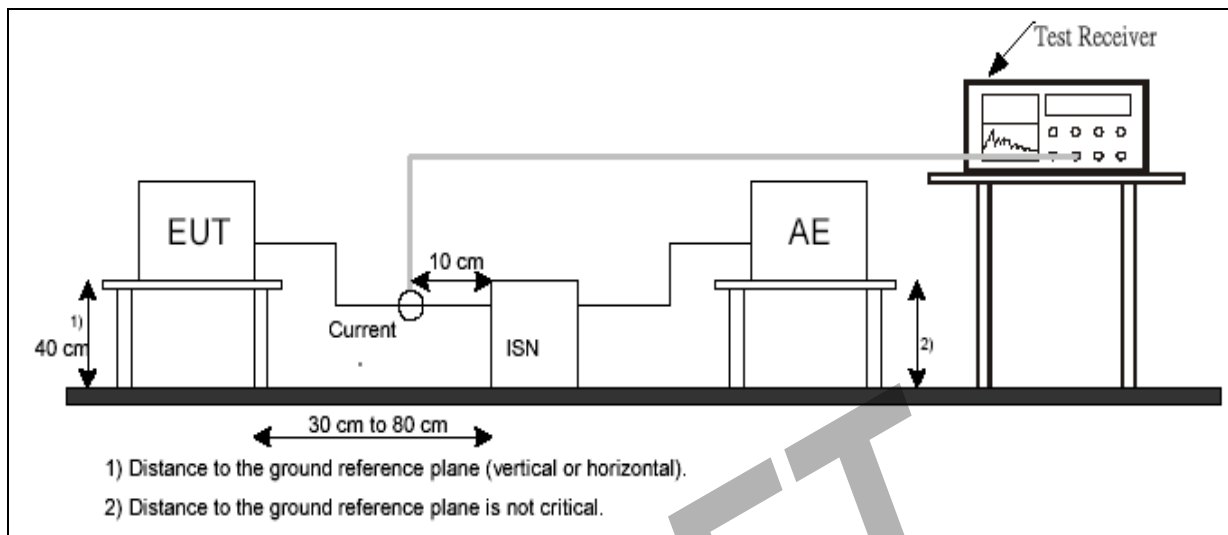
7.2.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-031)

- Selecting ISN for unscreened cable or a current probe for screened cable to take measurement.
- The port of the EUT was connected to the remote side support equipment through the ISN/Current Probe and communication in normal condition.
- Making a overall range scan by using the test receiver controlled by controller and record at least six highest emissions for showing in the test report.
- Emission frequency and amplitude were recorded into a computer in which correction factors were used to calculate the emission level and compare reading to the applicable limit.
- In case of measuring on the screened cable, the current limit shall be applied; otherwise the voltage limit should be applied.

Not applicable, the EUT doesn't have LAN Port or Modem port.

DRAFT

7.2.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

7.2.5. DATA SAMPLE

Freq. (MHz)	Reading (dBuV)	Factor (dB)	Result (dBuV)	Limit (dBuV)	Margin (dB)	Detector (P/Q/A)
x.xx	62.95	0.55	63.50	84	-20.50	Q

Freq. = Emission frequency in MHz
 Reading = Uncorrected Analyzer/Receiver reading
 Factor = Insertion loss of LISN + Cable Loss + Pulse Limit
 Result = Reading + Factor
 Limit = Limit stated in standard
 Margin = Reading in reference to limit
 P = Peak Reading
 Q = Quasi-peak Reading
 A = Average Reading

Calculation Formula

Margin (dB) = Result (dBuV) – Limit (dBuV)

Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 20 / 77
Rev.: 00

7.2.6. TEST RESULTS

Not applicable, the EUT doesn't have LAN Port or Modem port.

DRAFT

7.3. RADIATED EMISSION MEASUREMENT

7.3.1. LIMITS

Below 1GHz

FREQUENCY (MHz)	dBuV/m (At 10m)		dBuV/m (At 3m)	
	Class A	Class B	Class A	Class B
30 ~ 230	40	30	50	40
230 ~ 1000	47	37	57	47

Above 1GHz

Frequency (MHz)	Class A (dBuV/m) (At 3m)		Class B (dBuV/m) (At 3m)	
	Average	Peak	Average	Peak
1000 ~ 3000	56	76	50	70
3000 ~ 6000	60	80	54	74

NOTE: The lower limit shall apply at the transition frequencies.

According to EN 55032: 2015+A11: 2020 Table 1 the measurement frequency range is shown in the following table:

Table 1 – Required highest frequency for radiated measurement:

Highest internal frequency (F_X)	Highest internal frequency
$F_X \leq 108 \text{ MHz}$ $108 \text{ MHz} < F_X \leq 500 \text{ MHz}$ $500 \text{ MHz} < F_X \leq 1 \text{ GHz}$ $F_X > 1 \text{ GHz}$	1 GHz 2 GHz 5 GHz $5 \times F_X$ up to a maximum of 6 GHz
NOTE 1 For FM and TV broadcast receivers, F_X is determined from the highest frequency generated or used excluding the local oscillator and tuned frequencies.	
NOTE 2 F_X is defined in 3.1.19.	

Where F_X is unknown, the radiated emission measurements shall be performed up to 6GHz.

Radiated emissions from FM receivers:

Frequency range MHz	Measurement		Class B limit dB(μV/m)	
	Distance m	Detector type / bandwidth	Fundamental	Harmonics
			OATS / SAC (see Table A.1)	OATS / SAC (see Table A.1)
30 – 230	10	Quasi peak/ 120kHz	50	42
230 – 300				42
300 – 1000				46
30 – 230	3		60	52
230 – 300				52
300 – 1000				56

These relaxed limits apply only to emissions at the fundamental and harmonic frequencies of the local oscillator. Signals at all other frequencies shall be compliant with the limits given in 7.3.1 Class B Limit

7.3.2. TEST INSTRUMENTS

Below 1GHz

Wugu 10M Chamber					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
EMI Test Receiver	R&S	ESCI	100961	2024/04/29	2025/04/28
EMI Test Receiver	R&S	ESCI	100962	2024/05/08	2025/05/07
Pre-Amplifier	HP	8447D	2944A08150	2024/04/21	2025/04/20
Pre-Amplifier	HP	8447D	2944A07754	2024/04/21	2025/04/20
Bilog Antenna with 5dB Attenuator	TESEQ	CBL 6112D	31674	2024/02/19	2025/02/18
Bilog Antenna with 5dB Attenuator	TESEQ	CBL 6112D	31675	2024/03/08	2025/03/07
Cable	Huber Suhner	SUCOFLEX 104PEA	33948/4PEA	2024/04/21	2025/04/20
Cable	Huber Suhner	SUCOFLEX 104PEA	33949/4PEA	2024/04/21	2025/04/20
Cable	Huber Suhner	SUCOFLEX 104PEA	330029	2024/04/21	2025/04/20
Cable	Huber Suhner	SUCOFLEX 104PEA	24813	2024/04/21	2025/04/20
Turn Table	CCS	CC-T-1F	N/A	N.C.R	N.C.R
Controller	CCS	CC-C-1F	N/A	N.C.R	N.C.R
Antenna Tower	CCS	CC-A-1F	N/A	N.C.R	N.C.R
AC power source	APE	AFC-130	991259	N.C.R	N.C.R
Software	EZ-EMC (CCS-3A1RE)				

Note:

1. The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.
2. N.C.R. = No Calibration Required.

Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 23 / 77
Rev.: 00

Above 1GHz

Wugu 10M Chamber					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
Spectrum Analyzer	Agilent	E4446A	MY48250297	2024/07/18	2025/07/17
Cable	Huber Suhner	SUCOFLEX 104PEA	33945	2024/04/21	2025/04/20
Cable	Huber Suhner	SUCOFLEX 104PEA	329383	2024/04/21	2025/04/20
Pre-Amplifier	EMCI	EMC118A45SE	980820	2024/04/21	2025/04/20
Pre-Amplifier	EMCI	EMC184045SE	980872	2023/12/25	2024/12/24
Horn Antenna	RF SPIN	DRH18-E	210301A18ES	2024/01/24	2025/01/23
Horn Antenna	SCHWARZBECK	BBHA 9170	1134	2023/12/13	2024/12/12
Turn Table	CCS	CC-T-1F	N/A	N.C.R	N.C.R
Antenna Tower	Sunol Sciences	TLT2	031010-5	N.C.R	N.C.R
Controller	Sunol Sciences	SC104V	031010-1	N.C.R	N.C.R
AC power source	APE	AFC-130	991259	N.C.R	N.C.R
Software	EZ-EMC (CCS-3A1RE)				

Note:

1. The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.
2. N.C.R. = No Calibration Required.

7.3.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-031)

Procedure of Preliminary Test

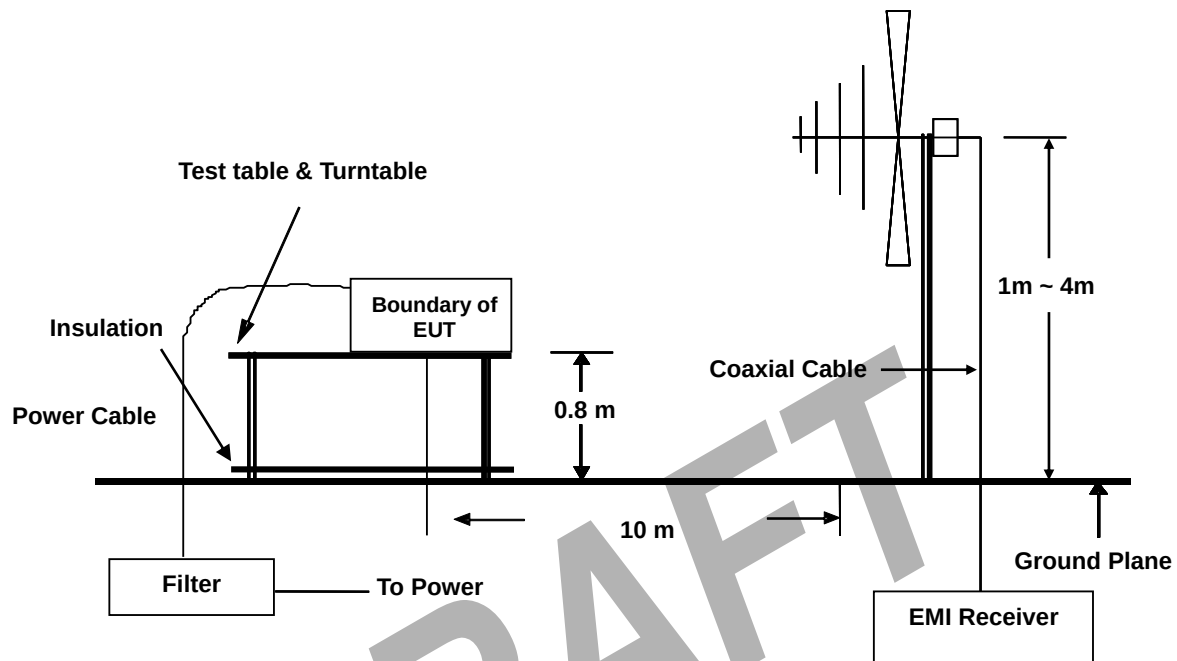
- The equipment was set up as per the test configuration to simulate typical usage per the user's manual. When the EUT is a tabletop system, a wooden turntable with a height of 0.8 meters is used which is placed on the ground plane. When the EUT is a floor standing equipment, it is placed on the ground plane which has a 15 cm non-conductive covering to insulate the EUT from the ground plane.
- Support equipment, if needed, was placed as per EN 55032.
- All I/O cables were positioned to simulate typical usage as per EN 55032.
- The EUT received AC power source from the outlet socket under the turntable. All support equipment power received from another socket under the turntable.
- The antenna was placed at 3 meters away from the EUT as stated in EN 55032. The antenna connected to the Spectrum Analyzer via a cable and at times a pre-amplifier would be used.
- The Analyzer / Receiver quickly scanned from 30MHz to 6000MHz. The EUT test program was started. Emissions were scanned and measured rotating the EUT to 360 degrees and positioning the antenna 1 to 4 meters above the ground plane, in both the vertical and the horizontal polarization, to maximize the emission reading level.
- The test mode(s) described in Item 4.1 were scanned during the preliminary test:
- After the preliminary scan, we found the test mode described in Item 4.1 producing the highest emission level.
- The EUT and cable configuration, antenna position, polarization and turntable position of the above highest emission level were recorded for the final test.

Procedure of Final Test

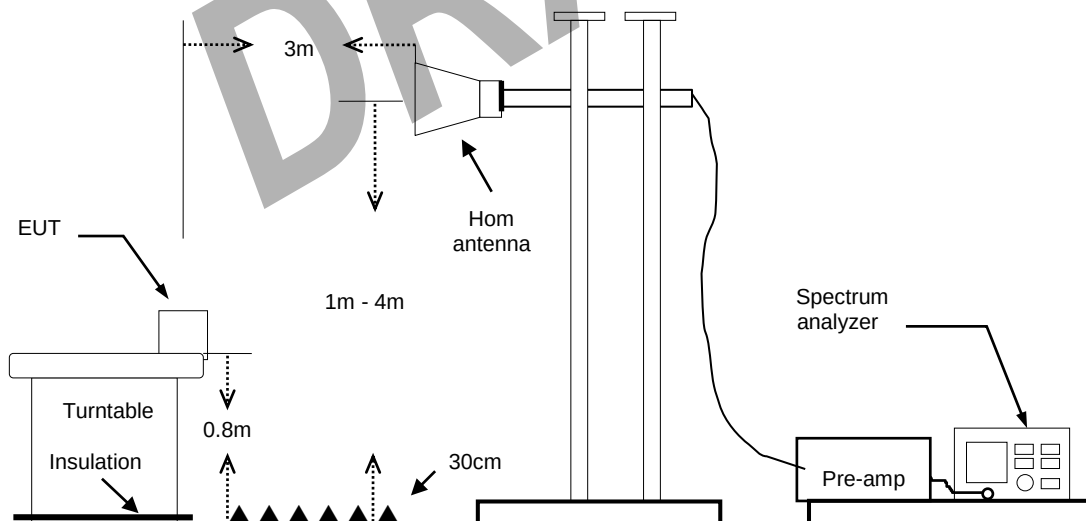
- EUT and support equipment were set up on the turntable as per the configuration with highest emission level in the preliminary test.
- The Analyzer / Receiver scanned from 30MHz to 6000MHz. Emissions were scanned and measured rotating the EUT to 360 degrees, varying cable placement and positioning the antenna 1 to 4 meters above the ground plane, in both the vertical and the horizontal polarization, to maximize the emission reading level.
- Recorded at least the six highest emissions. Emission frequency, amplitude, antenna position, polarization and turntable position were recorded into a computer in which correction factors were used to calculate the emission level and compare reading to the applicable limit. Below 1GHz the Q.P. reading and above 1GHz the Peak and Average reading are presented.
- The test data of the worst-case condition(s) was recorded.

7.3.4. TEST SETUP

Below 1GHz



Above 1GHz



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

7.3.5. DATA SAMPLE

Below 1GHz

Freq. (MHz)	Reading (dBuV)	Factor (dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector (P/Q)	Pol. (H/V)
x.xx	14.0	12.2	26.2	30	-3.8	Q	H

Above 1GHz

Freq. (MHz)	Reading (dBuV)	Factor (dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector (P/A)	Pol. (H/V)
x.xx	42.95	0.55	43.50	54	-10.50	A	H

Freq. = Emission frequency in MHz
 Reading = Uncorrected Analyzer/Receiver reading
 Factor = Antenna Factor + Cable Loss - Amplifier Gain
 Result = Reading + Factor
 Limit = Limit stated in standard
 Margin = Reading in reference to limit
 P = Peak Reading
 Q = Quasi-peak Reading
 A = Average Reading
 H = Antenna Polarization: Horizontal
 V = Antenna Polarization: Vertical

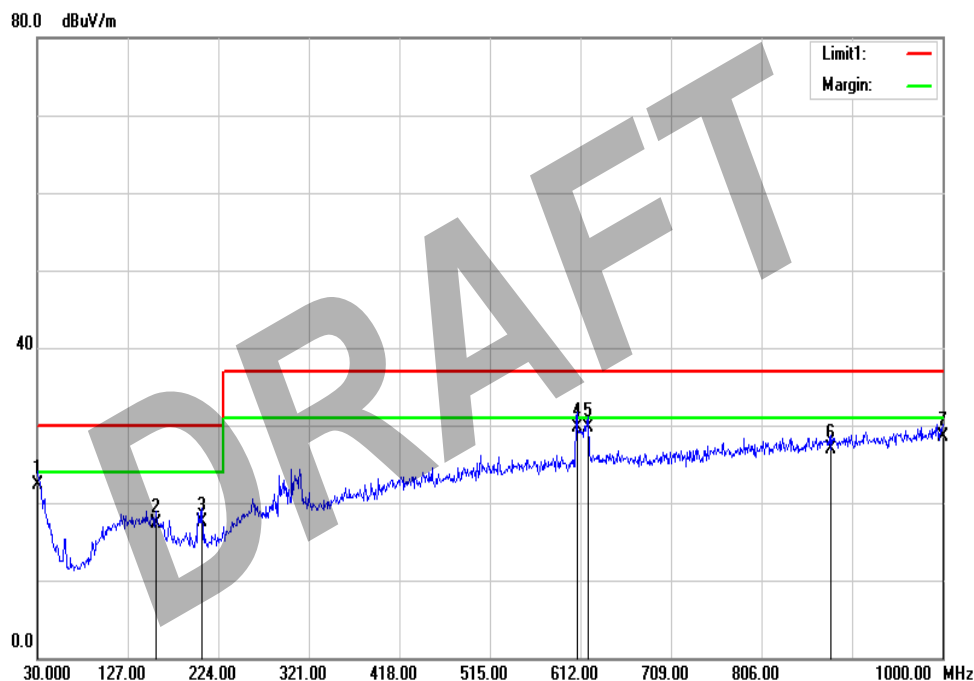
Calculation Formula

Margin (dB) = Result (dBuV/m) – Limit (dBuV/m)

7.3.6. TEST RESULTS

Below 1GHz

Model No.	DTTF1TU3A2V30AP	Test Mode	Mode 1
Environmental Conditions	25°C, 41% RH	Tested Date	2024/08/25
Antenna Pole	Vertical	Antenna Distance	10m
Detector Function	Quasi-peak.	Tested By	Rex Kuo
6dB Bandwidth	120 kHz		

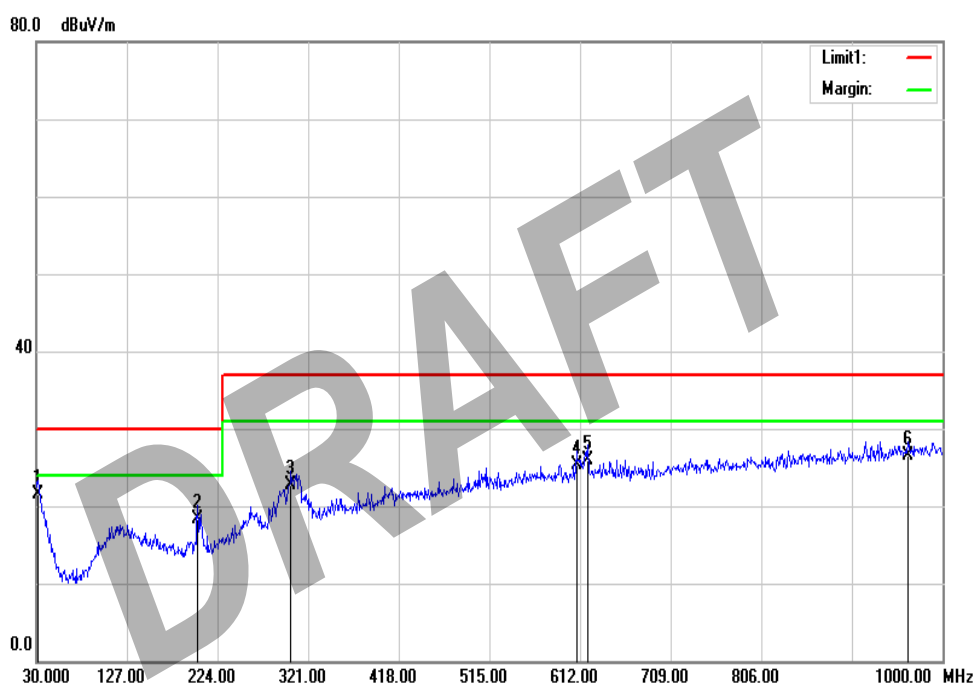


No.	Frequency (MHz)	Reading (dBuV)	Correction Factor (dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Height (cm)	Degree (°)	Remark
1	30.0000	24.76	-2.39	22.37	30.00	-7.63	300	359	QP
2	157.0700	26.30	-9.05	17.25	30.00	-12.75	100	277	QP
3	206.5400	26.97	-9.39	17.58	30.00	-12.42	200	260	QP
4	608.1200	28.49	1.30	29.79	37.00	-7.21	300	24	QP
5	619.7600	27.93	1.69	29.62	37.00	-7.38	323	0	QP
6	880.6900	22.01	4.80	26.81	37.00	-10.19	323	0	QP
7	1000.0000	22.13	6.37	28.50	37.00	-8.50	400	0	QP

Note:

1. P= Peak Reading; Q= Quasi-peak Reading.
2. The other emission levels were very low against the limit.

Model No.	DTTF1TU3A2V30AP	Test Mode	Mode 1
Environmental Conditions	25°C, 41% RH	Tested Date	2024/08/25
Antenna Pole	Horizontal	Antenna Distance	10m
Detector Function	Quasi-peak.	Tested By	Rex Kuo
6dB Bandwidth	120 kHz		



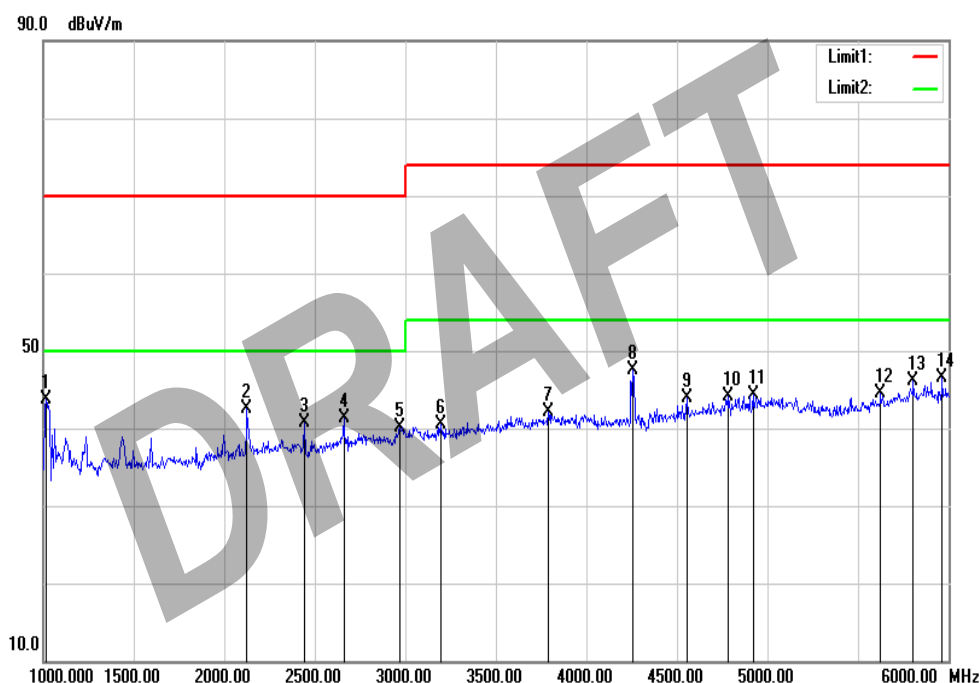
No.	Frequency (MHz)	Reading (dBuV)	Correction Factor (dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Height (cm)	Degree (°)	Remark
1	31.9400	25.26	-3.80	21.46	30.00	-8.54	300	0	QP
2	202.6600	28.53	-10.28	18.25	30.00	-11.75	400	292	QP
3	302.5700	28.41	-5.65	22.76	37.00	-14.24	200	0	QP
4	608.1200	25.66	-0.37	25.29	37.00	-11.71	103	0	QP
5	619.7600	26.35	-0.37	25.98	37.00	-11.02	100	53	QP
6	963.1400	22.45	4.08	26.53	37.00	-10.47	154	0	QP

Note:

1. P= Peak Reading; Q= Quasi-peak Reading.
2. The other emission levels were very low against the limit.

Above 1GHz

Model No.	DTTF1TU3A2V30AP	Test Mode	Mode 1
Environmental Conditions	25°C, 41% RH	Tested Date	2024/08/25
Antenna Pole	Vertical	Antenna Distance	3m
Highest frequency generated or used	208MHz	Upper frequency	2GHz
Detector Function	Peak	Tested By	Rex Kuo
6dB Bandwidth	1 MHz		



No.	Frequency (MHz)	Reading (dBuV)	Correction Factor (dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Height (cm)	Degree (°)	Remark
1	1015.000	52.40	-8.74	43.66	70.00	-26.34	233	360	peak
2	2127.500	47.62	-5.13	42.49	70.00	-27.51	200	245	peak
3	2442.500	45.42	-4.75	40.67	70.00	-29.33	100	105	peak
4	2660.000	45.07	-3.82	41.25	70.00	-28.75	100	359	peak
5	2970.000	42.93	-2.87	40.06	70.00	-29.94	300	95	peak
6	3195.000	43.31	-2.82	40.49	74.00	-33.51	124	0	peak
7	3790.000	42.45	-0.30	42.15	74.00	-31.85	200	0	peak
8	4255.000	47.86	-0.39	47.47	74.00	-26.53	100	341	peak

Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 30 / 77
Rev.: 00

No.	Frequency (MHz)	Reading (dBuV)	Correction Factor (dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Height (cm)	Degree (°)	Remark
9	4557.500	42.46	1.49	43.95	74.00	-30.05	300	123	peak
10	4782.500	42.45	1.59	44.04	74.00	-29.96	200	47	peak
11	4920.000	42.16	2.23	44.39	74.00	-29.61	400	152	peak
12	5625.000	41.57	2.88	44.45	74.00	-29.55	200	0	peak
13	5805.000	42.17	3.91	46.08	74.00	-27.92	100	227	peak
14	5965.000	42.70	3.74	46.44	74.00	-27.56	100	37	peak

REMARKS:

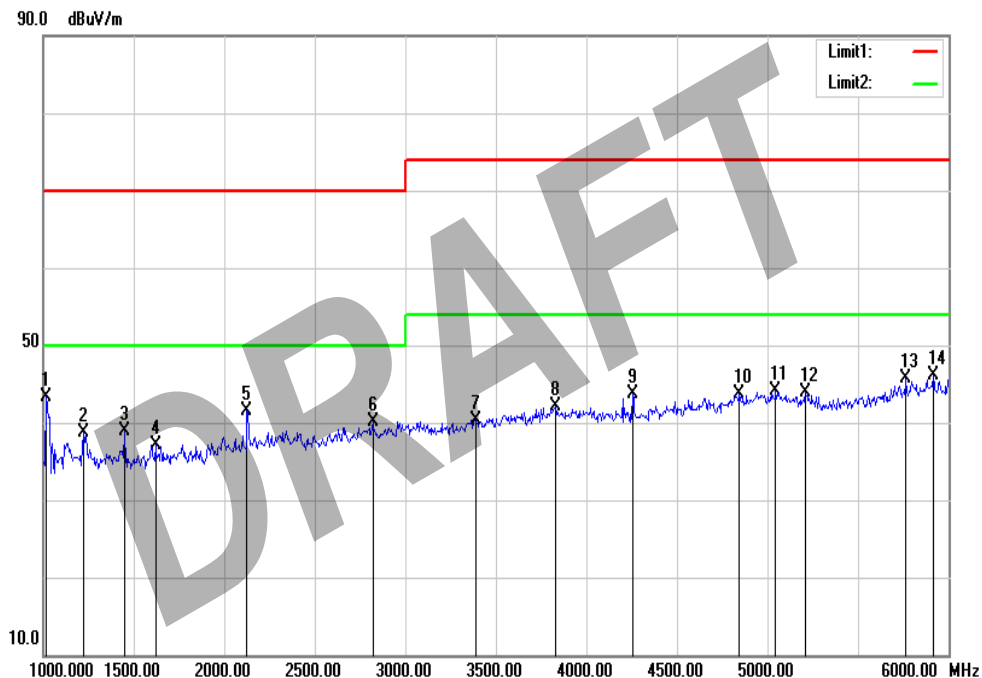
1. The other emission levels were very low against the limit.
2. Margin (dB) = Result (dBuV/m) – Limit (dBuV/m)

DRAFT

Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 31 / 77
Rev.: 00

Model No.	DTTF1TU3A2V30AP	Test Mode	Mode 1
Environmental Conditions	25°C, 41% RH	Tested Date	2024/08/25
Antenna Pole	Horizontal	Antenna Distance	3m
Highest frequency generated or used	208MHz	Upper frequency	2GHz
Detector Function	Peak	Tested By	Rex Kuo
6dB Bandwidth	1 MHz		



No.	Frequency (MHz)	Reading (dBuV)	Correction Factor (dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Height (cm)	Degree (°)	Remark
1	1012.500	51.94	-8.73	43.21	70.00	-26.79	200	284	peak
2	1222.500	46.69	-7.89	38.80	70.00	-31.20	195	360	peak
3	1447.500	46.30	-7.35	38.95	70.00	-31.05	100	103	peak
4	1620.000	44.33	-7.22	37.11	70.00	-32.89	200	86	peak
5	2125.000	46.70	-5.15	41.55	70.00	-28.45	100	360	peak
6	2820.000	43.31	-3.26	40.05	70.00	-29.95	300	122	peak
7	3387.500	42.35	-2.00	40.35	74.00	-33.65	300	217	peak
8	3827.500	42.26	-0.24	42.02	74.00	-31.98	300	18	peak

Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 32 / 77
Rev.: 00

No.	Frequency (MHz)	Reading (dBuV)	Correction Factor (dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Height (cm)	Degree (°)	Remark
9	4252.500	44.04	-0.40	43.64	74.00	-30.36	100	151	peak
10	4845.000	41.87	1.91	43.78	74.00	-30.22	300	141	peak
11	5042.500	41.84	2.30	44.14	74.00	-29.86	145	360	peak
12	5210.000	41.75	2.07	43.82	74.00	-30.18	300	273	peak
13	5762.500	41.94	3.77	45.71	74.00	-28.29	400	114	peak
14	5915.000	42.39	3.73	46.12	74.00	-27.88	300	0	peak

REMARKS:

1. The other emission levels were very low against the limit.
2. Margin (dB) = Result (dBuV/m) – Limit (dBuV/m)

DRAFT

7.4. CONDUCTED DIFFERENTIAL VOLTAGE EMISSIONS FROM CLASS B EQUIPMENT

Applicable to				
1. TV broadcast receiver tuner ports with an accessible connector				
2. RF modulator output ports				
3. FM broadcast receiver tuner ports with an accessible connector				
Frequency range MHz	Class B limits dB(μV) 75 Ω			Applicability
	other	Local Oscillator Fundamental	Local Oscillator Harmonics	
30 – 950	46	46	46	See a)
950 – 2 150	46	54	54	
950 – 2 150	46	54	54	See b)
30 – 300	46	54	50	See c)
300 – 1 000			52	
30 – 300	46	66	59	See d)
300 – 1 000			52	
30 – 950	46	76	46	See e)
950 – 2 150		n/a	54	
a) Television receivers (analogue or digital), video recorders and PC TV broadcast receiver tuner cards working in channels between 30 MHz and 1 GHz, and digital audio receivers.				
b) Tuner units (not the LNB) for satellite signal reception.				
c) Frequency modulation audio receivers and PC tuner cards.				
d) Frequency modulation car radios.				
e) Applicable to EUTs with RF modulator output ports (for example DVD equipment, video recorders, camcorders and decoders etc.) designed to connect to TV broadcast receiver tuner ports.				
Testing is required at only one EUT supply voltage and frequency.				
The term 'other' refers to all emissions other than the fundamental and the harmonics of the local oscillator.				
The test shall be performed with the device operating at each reception channel.				
The test shall cover the entire frequency range.				

7.4.1. TEST INSTRUMENTS

20 Immunity Room				
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Due
N/A				

7.4.2. TEST PROCEDURES (please refer to measurement standard or CCS SOP PA-041)

Procedure of Preliminary Test

- The equipment was set up as per the test configuration to simulate typical usage per the user's manual. The EUT was placed on a wooden table with a height of 0.8 meters was used that was placed on the ground plane.
- Support equipment, if needed, was placed as per EN 55032.
- All I/O cables were positioned to simulate typical usage as per EN 55032.
- The EUT received AC power source, from the outlet socket. All support equipment received power was from another socket.
- Added a 75 Ω →50 Ω matching network, between EUT and EMI test receiver to get impedance match condition during the test.
- The output level of the auxiliary signal generator shall be set to give the value of 60 dB (μ V) for FM receiver or 70 dB (μ V) for TV and VCR to the input of the frequency-modulation or television receiver (or video recorder) respectively, on a 75 Ω impedance. An additional amplifier should be inserted at the generator output, if necessary.
- The output level of the auxiliary signal generator shall be a standard TV color bar Move signal for TV receivers and video recorders with sound carrier that defined in Table A12 of EN 55032. An additional amplifier should be inserted at the generator output, if necessary.
- The results shall be expressed in the terms of the substitution voltage in decibels (μ V), as supplied by the standard signal generator. The specified source impedance of the receiver shall be stated with the results.
- When measurements are made at the antenna terminals of the EUT, an auxiliary signal generator shall be used to feed the equipment under test input with a standard test signal (see Table A.12 of EN 55032) at the receiver tuning frequency (30MHz to 2150MHz).
- The test mode(s) described in Item 4.1 were scanned during the preliminary test:
- After the preliminary scan, we found the test mode described in Item 4.1 producing the highest emission level.
- The EUT and cable configuration of the above highest emission levels were recorded for the final test.

Procedure of Final Test

- EUT and support equipment were set up on the table as per the configuration with highest emission level in the preliminary test.
- The Analyzer / Receiver scanned from 30MHz to 2150MHz. recorded the value, the local frequency, amplitude, were recorded in which correction factors were used to calculate the emission level and compare reading to the applicable limit, and only Q.P reading will record in this report.
- Recorded at least the six highest emissions. Emission frequencies, amplitude were recorded into a computer in which correction factors were used to calculate the emission level and compare reading to the applicable limit and only Q.P. reading is presented.
- The test data of the worst-case condition(s) was recorded.

7.4.3. DATA SAMPLE

Freq. (MHz)	Matching Factor (dB)	Spectrum Reading (dBuV)	SG Level (dBuV)	Emission (dBuV)	Limit Line (dBuV)	Over Limit (dB)	Note (F/H/O)
x.xx	12.2	14.0	38.4	26.2	46	-19.8	F

Freq. = Emission frequency in MHz
 Matching Factor = Matching network(50/75Ω) attenuation
 Spectrum Reading=Spectrum analyzer reading
 S.G. Level = Standard S.G. output level
 Emission = SG Level - Matching Factor
 Limit Line = Limit stated in standard
 Over Limit = Reading in reference to limit
 F = Fundamental
 H = Harmonics
 O = Other

Calculation Formula

Over Limit (dB) = Emission (dBuV) – Limit Line (dBuV)

7.4.4. TEST RESULTS

Not applicable, the EUT is not (TV broadcast receiver / RF modulator output ports / FM broadcast receiver).

7.5. HARMONICS CURRENT MEASUREMENT

7.5.1. LIMITS OF HARMONICS CURRENT MEASUREMENT

Limits for Class A equipment		Limits for Class D equipment		
Harmonics Order n	Max. permissible harmonics current A	Harmonics Order n	Max. permissible harmonics current per watt mA/W	Max. permissible harmonics current A
Odd harmonics		Odd Harmonics only		
3	2.30	3	3.4	2.30
5	1.14	5	1.9	1.14
7	0.77	7	1.0	0.77
9	0.40	9	0.5	0.40
11	0.33	11	0.35	0.33
13	0.21	13	0.30	0.21
15<=n<=39	0.15x15/n	15<=n<=39	3.85/n	0.15x15/n
Even harmonics				
2	1.08			
4	0.43			
6	0.30			
8<=n<=40	0.23x8/n			

Note:

1. Class A and Class D are classified according to item 7.5.3.
2. According to section 7 of EN IEC 61000-3-2, the above limits for all equipment except for lighting equipment having an active input power > 75 W and no limits apply for equipment with an active input power up to and including 75 W.

7.5.2. TEST INSTRUMENTS

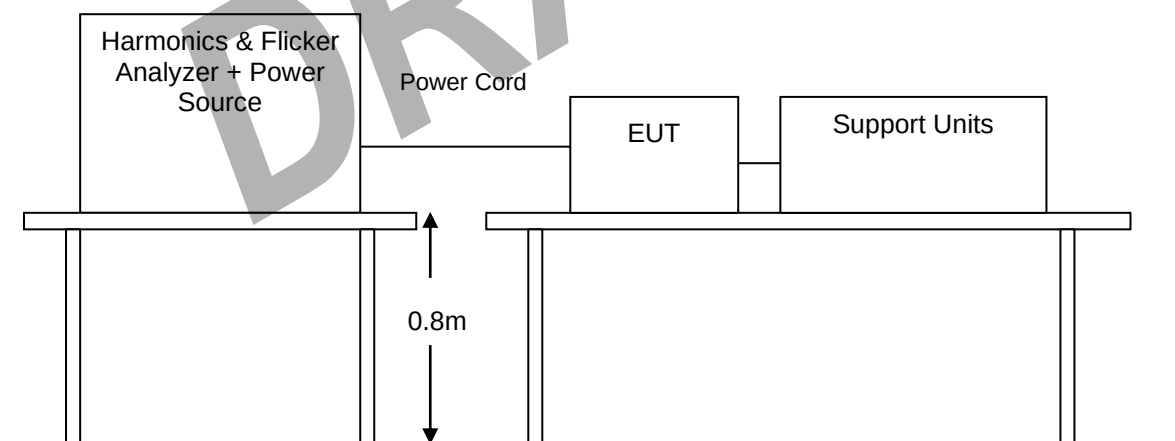
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
HARMONICS SYSTEM	EMC Partner	Harmonics-1000 /PS3/AFC-130	PS3-0211/107/9 91399	2024/06/04	2025/06/03
Software	HARCS Immunity (5.0)				

Note: The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.

7.5.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-029)

- The EUT was placed on the top of a wooden table 0.8 meters above the ground and operated to produce the maximum harmonic components under normal operating conditions for each successive harmonic component in turn.
- The classification of EUT is according to section 5 of EN IEC 61000-3-2.
- The EUT is classified as follows:
Class A: Balanced three-phase equipment, Household appliances excluding equipment as Class D, Tools excluding portable tools, Dimmers for incandescent lamps, audio equipment, equipment not specified in one of the three other classes.
Class B: Portable tools; Arc welding equipment which is not professional equipment.
Class C: Lighting equipment.
Class D: Equipment having a specified power less than or equal to 600 W of the following types: Personal computers and personal computer monitors and television receivers.
- The correspondent test program of test instrument to measure the current harmonics emanated from EUT is chosen. The measure time shall be not less than the time necessary for the EUT to be exercised.

7.5.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 38 / 77
Rev.: 00

7.5.5. TEST RESULTS

Test Mode 1

EUT max Power: 23.17W

Note: According to clause 7 of EN IEC 61000-3-2, equipment with a rated power of 75W or less, no limits apply.

DRAFT

7.6. VOLTAGE FLUCTUATION AND FLICKER MEASUREMENT

7.6.1. LIMITS OF VOLTAGE FLUCTUATION AND FLICKER MEASUREMENT

TEST ITEM	LIMIT	REMARK
P _{st}	1.0	P _{st} means short-term flicker indicator.
P _{lt}	0.65	P _{lt} means long-term flicker indicator.
T _{dt} (ms)	500	T _{dt} means maximum time that dt exceeds 3.3%.
d _{max} (%)	4%	d _{max} means maximum relative voltage change.
dc (%)	3.3%	dc means relative steady-state voltage change

7.6.2. TEST INSTRUMENTS

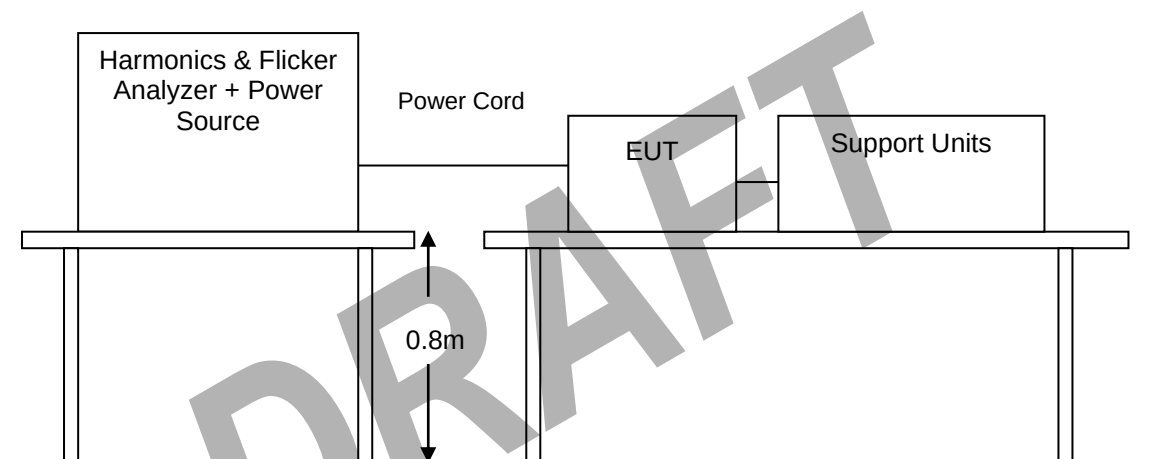
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
HARMONICS SYSTEM	EMC Partner	Harmonics-1000 /PS3/AFC-130	PS3-0211/107/9 91399	2024/06/04	2025/06/03
Software	HARCS Immunity (5.0)				

Note: The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.

7.6.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-030)

- The EUT was placed on the top of a wooden table 0.8 meters above the ground and operated to produce the most unfavorable sequence of voltage changes under normal operating conditions.
- During the flick measurement, the measure time shall include that part of whole operation cycle in which the EUT produce the most unfavorable sequence of voltage changes. The observation period for short-term flicker indicator is 10 minutes and the observation period for long-term flicker indicator is 2 hours.

7.6.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

7.6.5. TEST RESULTS

Observation Period (Tp)	10mins	Test Mode	Mode 1
Environmental Conditions	25°C, 59% RH	Tested By	Weiting Liu
Tested Date	2024/08/27		

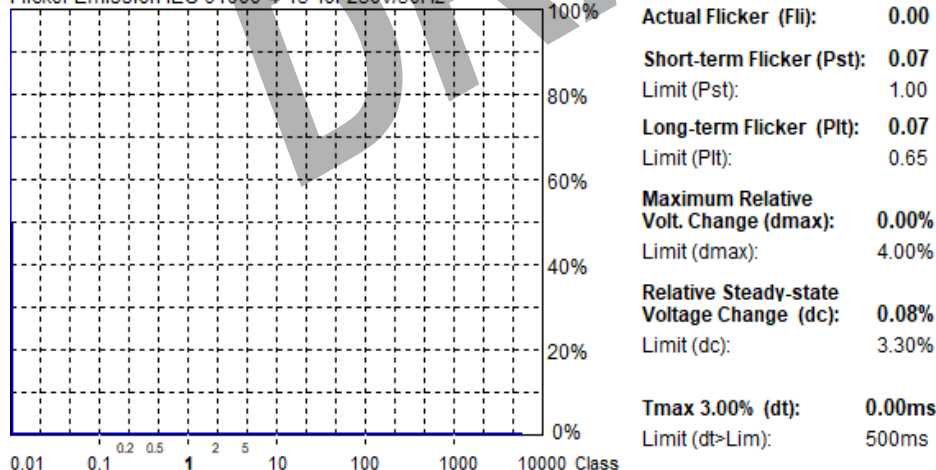
TEST PARAMETER	MEASUREMENT VALUE	LIMIT	REMARK
P _{st}	0.07	1.0	Pass
P _{lt}	0.07	0.65	Pass
T _{dt} (ms)	0.00	500	Pass
d _{max} (%)	0.00	4.0%	Pass
dc (%)	0.08	3.3%	Pass

Note: None.

Test result of EN 61000-3-3

Operator : Weiting Liu
EUT : Micro SD card
Model : DTTF1TU3A2V30AP
Remarks : Temp : 25°C Humidity : 59% RH

Flicker Emission IEC 61000-4-15 for 230V/50Hz



Flicker Emission - IEC 61000-3-3, EN 61000-3-3

Urms = 230.3 V P = 23.17 W
Irms = 0.266 A pf = 0.379

microSD card

Temp : 25°C Humidity : 59% RH

Test completed, Result: PASSED

2024/8/27 上午 01:41:12

Range: 2 A
V-nom: 231 V
TestTime: 10 min (100%)

HAR-1000 EMC-Partner

Full Bar: Actual Values
Empty Bar: Maximum Values
Circles: Average Values
Blue: Current, Green: Voltage, Red: Failed

8 IMMUNITY TEST

8.1. GENERAL DESCRIPTION

Product Standard	EN 55035: 2017+A11: 2020	
	Test Type	Minimum Requirement
Basic Standard, Specification, and Performance Criterion required	IEC 61000-4-2	Electrostatic Discharge – ESD: 8kV air discharge, 4kV Contact discharge, Performance Criterion B
	IEC 61000-4-3	Radio-Frequency Electromagnetic Field Susceptibility Test – RS: 80 ~1000 MHz, 1800, 2600, 3500, 5000MHz 3V/m, 80% AM(1kHz) Performance Criterion A
	IEC 61000-4-4	Electrical Fast Transient/Burst - EFT, AC Power Port: 1kV DC Power Port: 0.5kV Signal Ports and Telecommunication Ports: 0.5kV Performance Criterion B
	IEC 61000-4-5	Combination Wave for power port 1.2/50µs Open Circuit Voltage 8/20µs Short Circuit Current Combination Wave for Signal and Telecommunication port 10/700µs Open Circuit Voltage 5/320µs Short Circuit Current AC Power Port ~ line to line: 1kV, line to earth (ground): 2kV DC Power Port ~ line to earth: 0.5kV Performance Criterion B Signal and Telecommunication Ports ~ line to ground: 1kV Performance Criterion C
	IEC 61000-4-6	Conducted Radio Frequency Disturbances Test –CS: 0.15 ~ 10 MHz, 3Vrms, 10 ~ 30 MHz, 3 to 1 Vrms, 30 ~ 80 MHz, 1 Vrms, 80% AM, 1kHz, Performance Criterion A
	IEC 61000-4-8	Power frequency magnetic field immunity test 50 Hz & 60 Hz, 1A/m Performance Criterion A
	IEC 61000-4-11	Voltage Dips: i) >95% reduction for 0.5 periods, Performance Criterion B ii) 30% reduction for 25 periods, Performance Criterion C Voltage Interruptions: i) >95% reduction for 250 periods Performance Criterion C

8.2. GENERAL PERFORMANCE CRITERIA DESCRIPTION

Criteria A:	The apparatus shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus if used as intended.
Criteria B:	The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is however allowed. No change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus if used as intended.
Criteria C:	Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls.

8.3. ELECTROSTATIC DISCHARGE (ESD)

8.3.1. TEST SPECIFICATION

Basic Standard:	IEC 61000-4-2
Discharge Impedance:	330 ohm / 150 pF
Discharge Voltage:	Air Discharge: 2; 4; 8 kV (Direct) Contact Discharge: 2; 4 kV (Direct/Indirect)
Polarity:	Positive & Negative
Number of Discharge:	Air Discharge: min. 10 times at each test point for each polarity Contact Discharge: min. 200 times in total
Discharge Mode:	Single Discharge 1 second minimum

8.3.2. TEST INSTRUMENT

IMMUNITY SHIELDED ROOM					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
ESD Simulator	TESEQ	NSG 437	1539	2024/06/26	2025/06/25
Software	N/A				

Note: The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.

8.3.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-022)

The discharges shall be applied in two ways:

a) Contact discharges to the conductive surfaces and coupling planes:

The EUT shall be exposed to at least 200 discharges, 100 each at negative and positive polarity, at a minimum of four test points. One of the test points shall be subjected to at least 50 indirect discharges to the center of the front edge of the **Horizontal Coupling Plane (HCP)**. The remaining three test points shall each receive at least 50 direct contact discharges. If no direct contact test points are available, then at least 200 indirect discharges shall be applied in the indirect mode. Test shall be performed at a maximum repetition rate of one discharge per second.

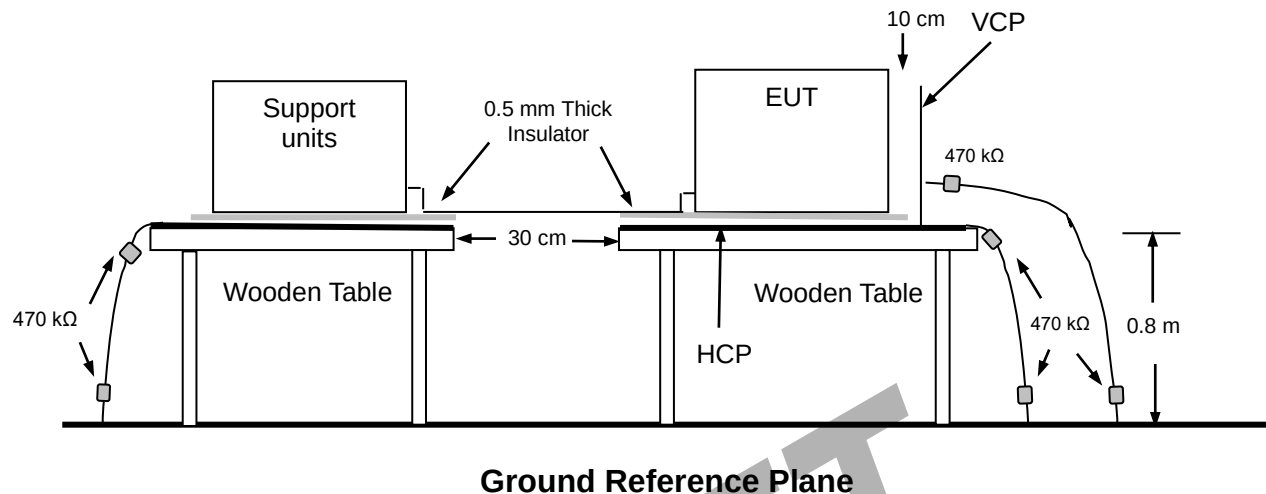
b) Air discharges at slots and apertures and insulating surfaces:

On those parts of the EUT where it is not possible to perform contact discharge testing, the equipment should be investigated to identify user accessible points where breakdown may occur. Such points are tested using the air discharge method. This investigation should be restricted to those area normally handled by the user. A minimum of 10 single air discharges shall be applied to the selected test point for each such area.

The basic test procedure was in accordance with IEC 61000-4-2:

- a) The EUT was located 0.1 m minimum from all side of the **HCP** (dimensions 1.6m x 0.8m).
- b) The support units were located another table 30 cm away from the EUT, but direct support unit was/were located at same location as EUT on the HCP and keep at a distance of 10 cm with EUT.
- c) The time interval between two successive single discharges was at least 1 second.
- d) Contact discharges were applied to the non-insulating coating, with the pointed tip of the generator penetrating the coating and contacting the conducting substrate.
- e) Air discharges were applied with the round discharge tip of the discharge electrode approaching the EUT as fast as possible (without causing mechanical damage) to touch the EUT. After each discharge, the ESD generator was removed from the EUT and re-triggered for a new single discharge. The test was repeated until all discharges were complete.
- f) At least ten single discharges (in the most sensitive polarity) were applied at the front edge of each **HCP** opposite the center point of each unit of the EUT and 0.1 meters from the front of the EUT. The long axis of the discharge electrode was in the plane of the **HCP** and perpendicular to its front edge during the discharge.
- g) At least ten single discharges (in the most sensitive polarity) were applied to the center of one vertical edge of the **Vertical Coupling Plane (VCP)** in sufficiently different positions that the four faces of the EUT were completely illuminated. The **VCP** (dimensions 0.5m x 0.5m) was placed vertically to and 0.1 meters from the EUT.

8.3.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

NOTE:

TABLE-TOP EQUIPMENT

The configuration consisted of a wooden table 0.8 meters high standing on the **Ground Reference Plane**. The **GRP** consisted of a sheet of aluminum at least 0.25mm thick, and 2.5 meters square connected to the protective grounding system. A **Horizontal Coupling Plane** (1.6m x 0.8m) was placed on the table and attached to the **GRP** by means of a cable with 940k total impedance. The equipment under test, was installed in a representative system as described in section 7 of IEC 61000-4-2, and its cables were placed on the **HCP** and isolated by an insulating support of 0.5mm thickness. A distance of 1-meter minimum was provided between the EUT and the walls of the laboratory and any other metallic structure.

FLOOR-STANDING EQUIPMENT

The equipment under test was installed in a representative system as described in section 7 of IEC 61000-4-2, and its cables were isolated from the Ground Reference Plane by an insulating support of 0.1-meter thickness. The GRP consisted of a sheet of aluminum that is at least 0.25mm thick, and 2.5 meters square connected to the protective grounding system and extended at least 0.5 meters from the EUT on all sides.

8.3.5. TEST RESULTS

Temperature	23°C	Humidity	50% RH
Pressure	1001mbar	Tested By	Eason Liu
Tested Date	2024/09/03	Test Mode	Mode 1
Required Passing Performance	Criterion B		

Air Discharge												
Test Points	Test Levels							Results				
	± 2 kV	Performance Criterion		± 4 kV	Performance Criterion		± 8 kV	Performance Criterion		Pass	Fail	Observation
Front	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Back	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Left	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Right	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Top	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Bottom	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2

Contact Discharge									
Test Points	Test Levels						Results		
	± 2 kV	Performance Criterion		± 4 kV	Performance Criterion		Pass	Fail	Observation
Front	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Back	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Left	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Right	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Top	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2
Bottom	☐	☐ A	☐ B	☐	☐ A	☐ B	☐	☐	Note 2

Discharge To Horizontal Coupling Plane						
Side of EUT	Test Levels		Results			
	± 2 kV	± 4 kV	Pass	Fail	Performance Criterion	Observation
Front	☒	☒	☒	☐	☒ A ☐ B	Note 1
Back	☒	☒	☒	☐	☒ A ☐ B	Note 1
Left	☒	☒	☒	☐	☒ A ☐ B	Note 1
Right	☒	☒	☒	☐	☒ A ☐ B	Note 1

Discharge To Vertical Coupling Plane						
Side of EUT	Test Levels		Results			
	± 2 kV	± 4 kV	Pass	Fail	Performance Criterion	Observation
Front	☒	☒	☒	☐	☒ A ☐ B	Note 1
Back	☒	☒	☒	☐	☒ A ☐ B	Note 1
Left	☒	☒	☒	☐	☒ A ☐ B	Note 1
Right	☒	☒	☒	☐	☒ A ☐ B	Note 1

Note:

1. There was no change compared with initial operation during the test.
2. Means that no discharge point had been occurred during that particular coupling method.

8.4. RADIATED, RADIO-FREQUENCY, ELECTROMAGNETIC FIELD (RS)

8.4.1. TEST SPECIFICATION

Basic Standard: IEC 61000-4-3

Frequency Range: 80 MHz ~ 1000 MHz
1800 MHz, 2600 MHz, 3500 MHz, 5000MHz (+/-1%)

Field Strength: 3 V/m

Modulation: 1kHz sine Wave, 80%, AM Modulation.

Frequency Step: 1 % of the preceding frequency

Polarity of Antenna: Horizontal and Vertical

Test Distance: 3 m

Antenna Height: 1.5 m

8.4.2. TEST INSTRUMENT

RS Chamber					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
Signal Generator	R&S	SMJ100A	101258	2024/06/22	2025/06/21
Power Sensor	R&S	NRP-Z21	101860	2024/06/15	2025/06/14
Power Sensor	R&S	NRP-Z21	101861	2024/06/15	2025/06/14
Controller	AR	SC1000M3	0433953	N.C.R	N.C.R
Power Amplifier	AR	250W1000BM1	0579919	N.C.R	N.C.R
Power Amplifier	AR	50S1G6M1	0433952	N.C.R	N.C.R
Horn Antenna	SCHWARZBECK	STLP 9149	9149-261	N.C.R	N.C.R
Dual Directional Coupler	RD Microswaves	C1-A47NFNF	31	N.C.R	N.C.R
Dual Directional Coupler	AR	DC6180A	433803	N.C.R	N.C.R
Antenna	AR	ATL80M1G	044851	N.C.R	N.C.R
Software	EMC 32(V10.20.01)				

Note:

1. The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.
2. N.C.R. = No Calibration required.

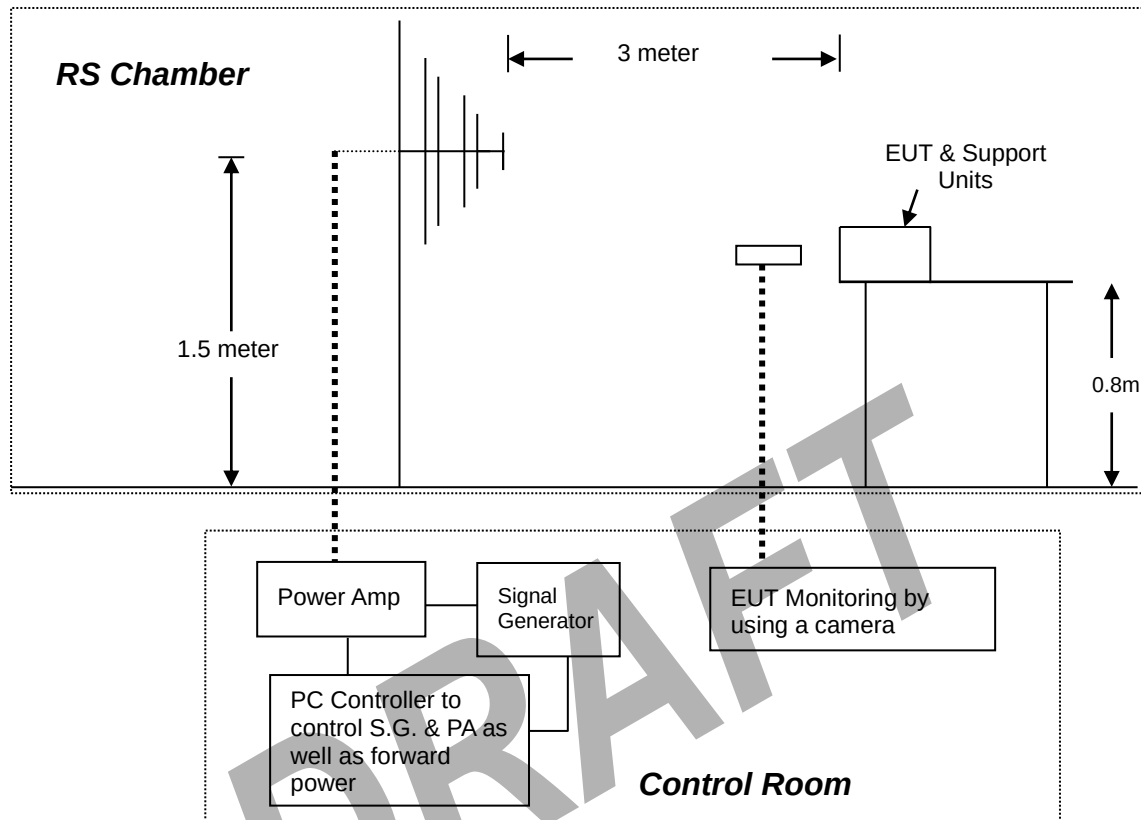
8.4.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-023)

The test procedure was in accordance with IEC 61000-4-3

- a) The testing was performed in a fully anechoic chamber. The transmit antenna was located at a distance of 3 meters from the EUT.
- b) The frequency range is swept from 80 MHz to 1000 MHz, 1800, 2600, 3500, and 5000 MHz with the signal 80% amplitude modulated with a 1 kHz sine-wave. The rate of sweep did not exceed 1.5×10^{-3} decade/s, where the frequency range is swept incrementally, the step size was 1% of preceding frequency value.
- c) The dwell time at each frequency shall be not less than the time necessary for the EUT to be able to respond.
- d) The field strength level was 3V/m.
- e) The test was performed with the EUT exposed to both vertically and horizontally polarized fields on each of the four sides

DRAFT

8.4.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

NOTE:

TABLETOP EQUIPMENT

The EUT installed in a representative system as described in section 7 of IEC 61000-4-3 was placed on a non-conductive table 0.8 meters in height. The system under test was connected to the power and signal wire according to relevant installation instructions.

FLOOR STANDING EQUIPMENT

The EUT installed in a representative system as described in section 7 of IEC 61000-4-3 was placed on a non-conductive wood support 0.1 meters in height. The system under test was connected to the power and signal wire according to relevant installation instructions.

8.4.5. TEST RESULTS

Temperature	23°C	Humidity	50% RH
Pressure	1001mbar	Dwell Time	3 sec.
Tested Date	2024/09/03	Tested By	Eason Liu
Test Mode	Mode 1	Required Passing Performance	Criterion A

Frequency (MHz)	Polarity	Azimuth	Field Strength (V/m)	Performance Criterion	Observation	Result
80 ~ 1000	V&H	0	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
80 ~ 1000	V&H	90	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
80 ~ 1000	V&H	180	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
80 ~ 1000	V&H	270	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
1800	V&H	0	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
1800	V&H	90	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
1800	V&H	180	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
1800	V&H	270	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
2600	V&H	0	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
2600	V&H	90	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
2600	V&H	180	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
2600	V&H	270	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
3500	V&H	0	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
3500	V&H	90	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
3500	V&H	180	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
3500	V&H	270	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
5000	V&H	0	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
5000	V&H	90	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
5000	V&H	180	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
5000	V&H	270	3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS

Note: 1. There was no change compared with the initial operation during the test.

8.5. ELECTRICAL FAST TRANSIENT (EFT)

8.5.1. TEST SPECIFICATION

Basic Standard: IEC 61000-4-4

Test Voltage: AC Power Port: 1kV
DC Power Port: 0.5kV
Signal Ports and Telecommunication Ports: 0.5kV

Polarity: Positive & Negative

Impulse Frequency: 5 kHz

Impulse Wave-shape: 5/50 ns

Burst Duration: 15 ms

Burst Period: 300 ms

Test Duration: Not less than 1 min.

8.5.2. TEST INSTRUMENT

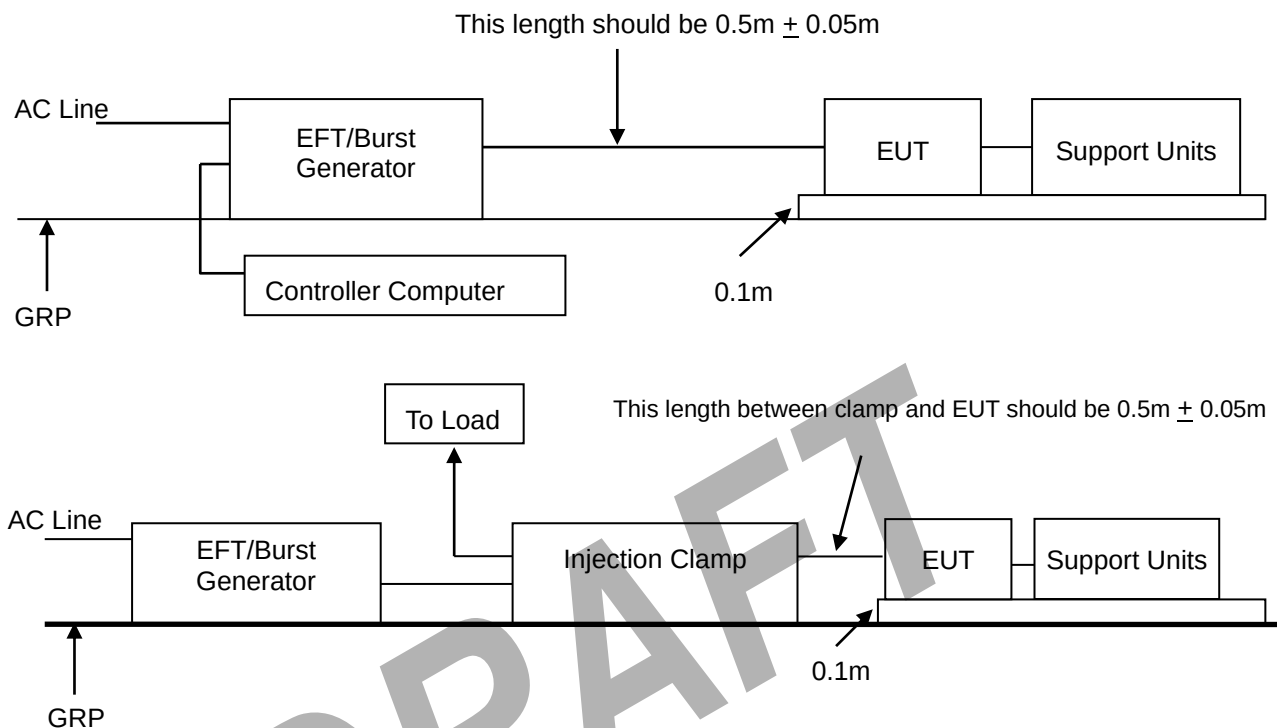
Immunity Shield Room					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
EMC Immunity Tester	EMC Partner	TRA2006	1144	2023/10/06	2024/10/05
CDN	EMC Partner	CDN-UTP8 ED3	046	2023/10/06	2024/10/05
Clamp	EMC Partner	CN-EFT1000	683	2024/08/02	2025/08/01
Software	N/A				

Note: The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.

8.5.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-024)

- Both positive and negative polarity discharges were applied.
- The length of the “hot wire” from the coaxial output of the EFT generator to the terminals on the EUT should not exceed 0.5 meter.
- The duration time of each test sequential was 1 minute.
- The transient/burst waveform was in accordance with IEC 61000-4-4 5/50ns

8.5.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

NOTE:

TABLETOP EQUIPMENT

The configuration consisted of a wooden table (0.1m high) standing on the Ground Reference Plane. The GRP consisted of a sheet of aluminum (at least 0.25mm thick and 2.5m square) connected to the protective grounding system. A minimum distance of 0.5m was provided between the EUT and the walls of the laboratory or any other metallic structure.

FLOOR STANDING EQUIPMENT

The EUT installed in a representative system as described in section 7 of IEC 61000-4-4 and its cables were isolated from the Ground Reference Plane by an insulating support that is 0.1-meter thick. The GRP consisted of a sheet of aluminum (at least 0.25mm thick and 2.5m square) connected to the protective grounding system.



Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 54 / 77
Rev.: 00

8.5.5. TEST RESULTS

Temperature	23°C	Humidity	50% RH
Pressure	1001mbar	Tested By	Eason Liu
Tested Date	2024/09/03	Test Mode	Mode 1
Required Passing Performance	Criterion B		

Test Point	Polarity	Test Level (kV)	Performance Criterion	Observation	Result
L	+/-	1	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
N	+/-	1	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
PE	+/-	1	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
L + N	+/-	1	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
L + PE	+/-	1	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
N + PE	+/-	1	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
L + N + PE	+/-	1	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS

Note: 1. There was no change compared with the initial operation during the test.

8.6. SURGE IMMUNITY TEST

8.6.1. TEST SPECIFICATION

Basic Standard: IEC 61000-4-5

Wave-Shape: Combination Wave
1.2/50 μ s Open Circuit Voltage
8/20 μ s Short Circuit Current

Combination Wave for Signal and Telecommunication port
10/700 μ s Open Circuit Voltage
5/320 μ s Short Circuit Current

Test Voltage: AC Power Port ~ line to line: 1kV
line to earth (ground): 2kV
DC Power Port ~ line to earth: 0.5kV
Performance Criterion B
Signal and Telecommunication Ports ~ line to ground: 1kV
Performance Criterion C

Surge Input/Output: AC Power Line: L - N

Generator Source Impedance: 2 ohm between networks
12 ohm between network and ground

Polarity: Positive/Negative

Phase Angle: 0° / 90° / 180° / 270°

Pulse Repetition Rate: 1 time / min. (maximum)

Number of Tests: 5 positive and 5 negative at selected points

8.6.2. TEST INSTRUMENT

Immunity Shield Room					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
EMC Immunity Tester	EMC Partner	TRA2006	1144	2023/10/06	2024/10/05
CDN	EMC Partner	CDN-UTP8 ED3	046	2023/10/06	2024/10/05
Clamp	EMC Partner	CN-EFT1000	683	2024/08/02	2025/08/01
Software	N/A				

Note: The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.

8.6.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-025)

a) For EUT power supply:

The surge is applied to the EUT power supply terminals via the capacitive coupling network. Decoupling networks are required in order to avoid possible adverse effects on equipment not under test that may be powered by the same lines, and to provide sufficient decoupling impedance to the surge wave. The power cord between the EUT and the coupling/decoupling networks was shorter than 2 meters in length.

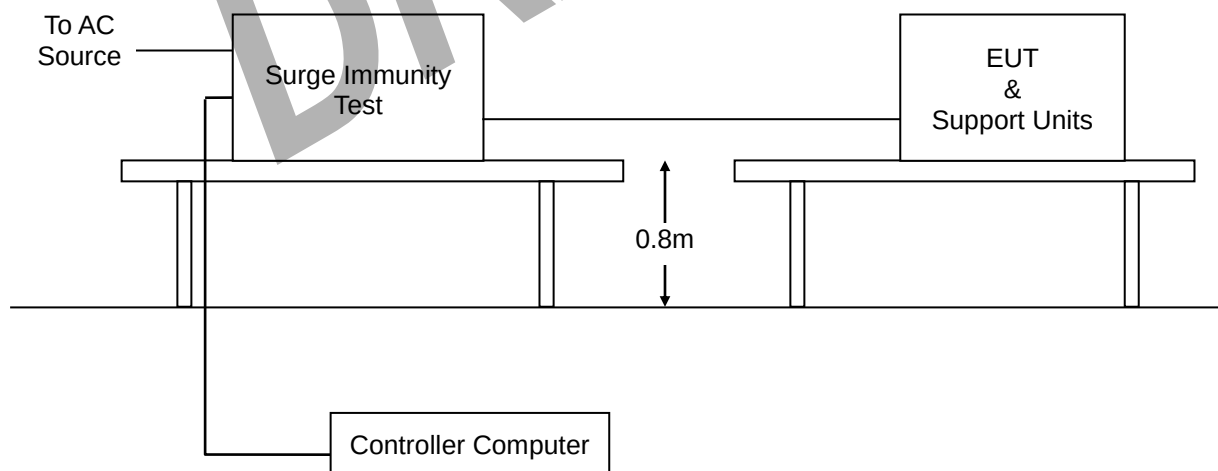
b) For test applied to unshielded un-symmetrically operated interconnection lines of EUT:

The surge was applied to the lines via the capacitive coupling. The coupling / decoupling networks didn't influence the specified functional conditions of the EUT. The interconnection line between the EUT and the coupling/decoupling networks was shorter than 2 meters in length.

c) For test applied to unshielded symmetrically operated interconnection / telecommunication lines of EUT:

The surge was applied to the lines via gas arrestors coupling. Test levels below the ignition point of the coupling arrestor were not specified. The interconnection line between the EUT and the coupling/decoupling networks was shorter than 2 meters in length.

8.6.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.



Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 57 / 77
Rev.: 00

8.6.5. TEST RESULT

Temperature	23°C	Humidity	50% RH
Pressure	1001mbar	Tested By	Eason Liu
Tested Date	2024/09/03	Test Mode	Mode 1
Required Passing Performance	Criterion B		

Test Point	Polarity	Test Level (kV)	Performance Criterion	Observation	Result
L – N	+/-	1	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
L – PE	+/-	2	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
N – PE	+/-	2	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS

Note: 1. There was no change compared with initial operation during the test.

DRAFT

8.7. CONDUCTED RADIO FREQUENCY DISTURBANCES (CS)

8.7.1. TEST SPECIFICATION

Basic Standard:	IEC 61000-4-6
Frequency Range:	0.15 ~ 10 MHz 10 ~ 30 MHz 30 ~ 80 MHz
Field Strength:	3 Vrms 3 to 1 Vrms 1 Vrms
Modulation:	1kHz Sine Wave, 80%, AM Modulation
Frequency Step:	1 % of the preceding frequency
Coupled cable:	Power Mains, Unshielded
Coupling device:	☐ CDN-M2 (2 wires) ☒ CDN-M3 (3 wires) ☐ CDN-T8 for LAN ☐ Clamp

8.7.2. TEST INSTRUMENT

CS Room					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
CDN	LUTHI	CDN M2/M3	P1325119693	2024/08/12	2025/08/11
CDN	LUTHI	CDN T4-3A	P1339124699	2024/08/12	2025/08/11
EM Injection Clamp	FCC	F-203I-23mm	421	2024/08/03	2025/08/02
CDN	FCC	FCC-801-M2-16A	121695	2024/08/12	2025/08/11
CDN	TESEQ	CDN T8-10	40374	2024/08/12	2025/08/11
Immunity Test System	TESEQ	NSG 4070C-35	60389	2023/11/17	2024/11/16
Software	NSG 4070 control program(V 1.4.0)				

Note:

- The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.
- N.C.R.= No Calibration required.

8.7.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-026)

The EUT shall be tested within its intended operating and climatic conditions.

The test shall be performed with the test generator connected to each of the coupling and decoupling devices in turn, while the other non-excited RF input ports of the coupling devices are terminated by a 50-ohm load resistor.

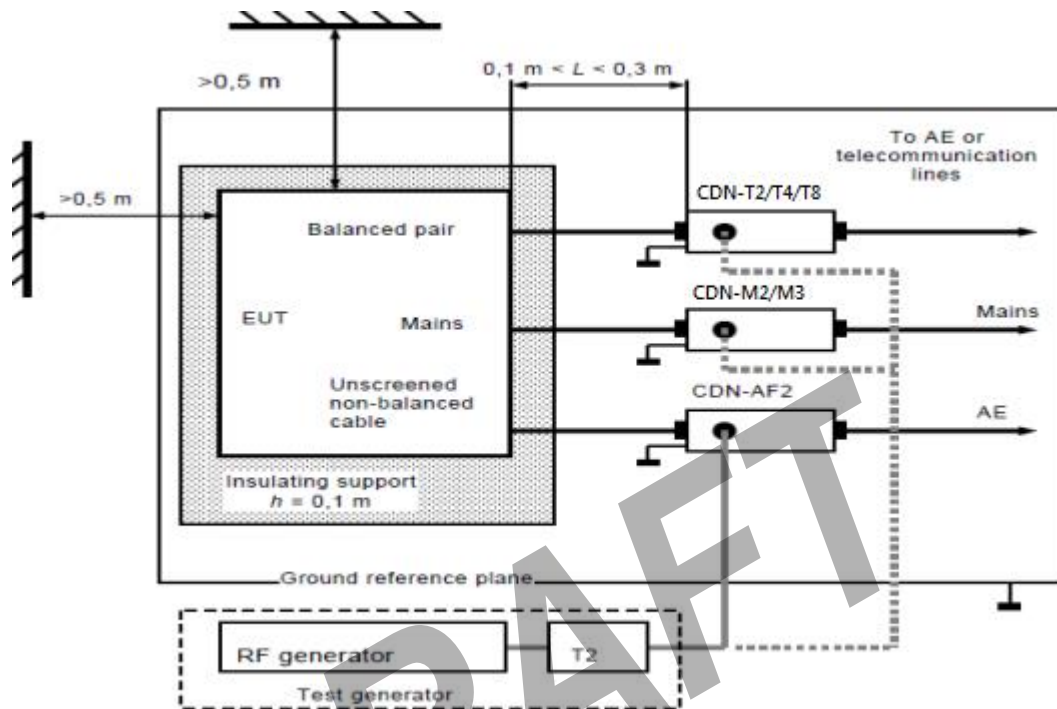
The frequency range was swept from 150 kHz to 80 MHz, using the signal level established during the setting process and with a disturbance signal of 80 % amplitude. The signal was modulated with a 1 kHz sine wave, pausing to adjust the RF signal level or the switch coupling devices as necessary. The sweep rate was 1.5×10^{-3} decades/s. Where the frequency range is swept incrementally, the step size was 1 % of preceding frequency value from 150 kHz to 80 MHz.

The dwell time at each frequency was less than the time necessary for the EUT to be exercised, and able to respond. Sensitive frequencies such as clock frequency(ies) and harmonics or frequencies of dominant interest, was analyzed separately.

Attempts were made to fully exercise the EUT during testing, and to fully interrogate all exercise modes selected for susceptibility.

DRAFT

8.7.4. TEST SETUP



Note:

1. The EUT is setup 0.1m above Ground Reference Plane
2. The CDNs and / or EM clamp used for real test depend on ports and cables configuration of EUT.

- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

NOTE:

TABLE-TOP AND FLOOR-STANDING EQUIPMENT

The equipment to be tested is placed on an insulating support of 0.1 meters height above a ground reference plane. All relevant cables shall be provided with the appropriate coupling and decoupling devices at a distance between 0.1 meters and 0.3 meters from the projected geometry of the EUT on the ground reference plane.



Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 61 / 77
Rev.: 00

8.7.5. TEST RESULTS

Temperature	23°C	Humidity	50% RH
Pressure	1001mbar	Tested By	Eason Liu
Tested Date	2024/09/03	Test Mode	Mode 1
Required Passing Performance	Criterion A		

Frequency Band (MHz)	Field Strength (Vrms)	Cable	Injection Method	Performance Criterion	Observation	Result
0.15 ~ 10	3	AC Power Line	CDN-M3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
10 ~ 30	3 to 1	AC Power Line	CDN-M3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS
30 ~ 80	1	AC Power Line	CDN-M3	☒ A ☐ B	Note ☒ 1 ☐ 2	PASS

Note: 1. There was no change compared with initial operation during the test.

DRAFT

8.8. POWER FREQUENCY MAGNETIC FIELD

8.8.1. TEST SPECIFICATION

Basic Standard: IEC 61000-4-8

Frequency Range: 50Hz & 60Hz

Field Strength: 1 A/m

Observation Time: 1 minute

Inductance Coil: Rectangular type, 1mx1m

8.8.2. TEST INSTRUMENT

Immunity Shield Room					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
Triax Elf Magnetic Field Meter	F.W.BELL	4090	9711	2024/06/13	2025/06/12
Magnetic field Tester	HAEFFLY	MAG 100.1	081436-02	N.C.R	N.C.R
Software	N/A				

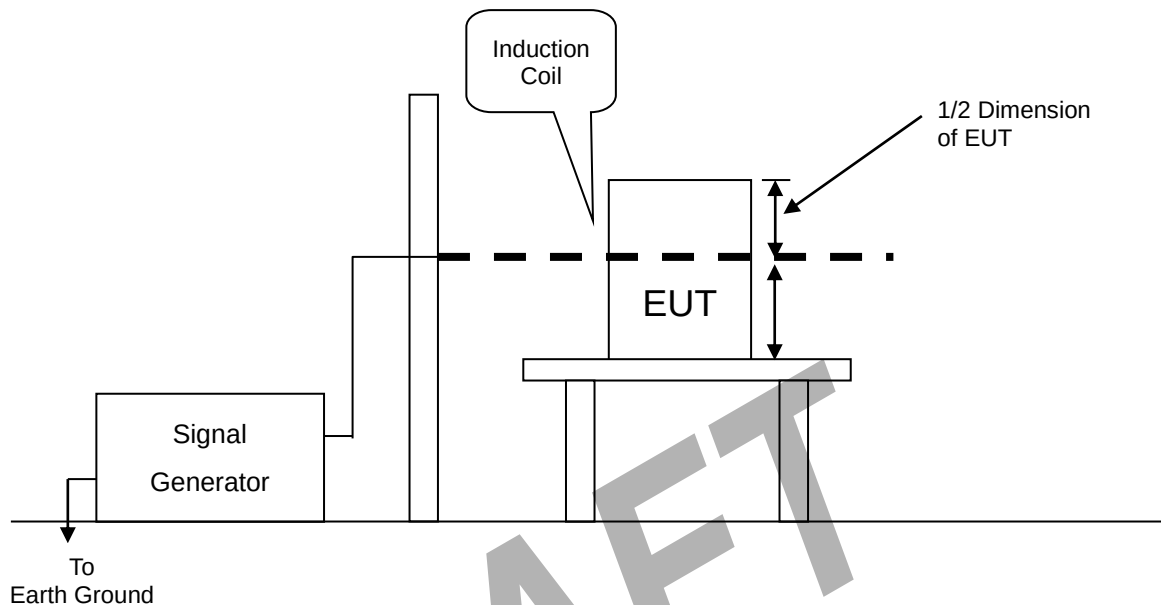
Note:

1. The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.
2. N.C.R. = No Calibration Required.

8.8.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-027)

- a. The equipment is configured and connected to satisfy its functional requirements. It shall be placed on the GRP with the interposition of a 0.1m-thick insulating support.
- b. The equipment cabinets shall be connected to the safety earth directly on the GRP via the earth terminal of the EUT.
- c. The power supply, input and output circuits shall be connected to the sources of power supply, control and signal.
- d. The cables supplied or recommended by the equipment manufacturer shall be used. 1 meter of all cables used shall be exposed to the magnetic field.

8.8.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.

NOTE:

TABLETOP EQUIPMENT

The equipment shall be subjected to the test magnetic field by using the induction coil of standard dimension (1 m x 1 m). The induction coil shall then be rotated by 90 degrees in order to expose the EUT to the test field with different orientations.

FLOOR-STANDING EQUIPMENT

The equipment shall be subjected to the test magnetic field by using induction coils of suitable dimensions. The test shall be repeated by moving and shifting the induction coils, in order to test the whole volume of the EUT for each orthogonal direction. The test shall be repeated with the coil shifted to different positions along the side of the EUT, in steps corresponding to 50 % of the shortest side of the coil. The induction coil shall then be rotated by 90 degrees in order to expose the EUT to the test field with different orientations.



Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 64 / 77
Rev.: 00

8.8.5. TEST RESULTS

Temperature	23°C	Humidity	50% RH
Pressure	1001mbar	Tested By	Eason Liu
Tested Date	2024/09/03	Test Mode	Mode 1
Required Passing Performance	Criterion A		

Direction	Field Strength (A/m)		Performance Criterion of testing	Observation	Results
	50Hz	60Hz			
X	1	1	☒ A ☐ B	Note 1	Pass
Y	1	1	☒ A ☐ B	Note 1	Pass
Z	1	1	☒ A ☐ B	Note 1	Pass

Note: 1. There was no change compared with the initial operation during the test.

8.9. VOLTAGE DIP & VOLTAGE INTERRUPTIONS

8.9.1. TEST SPECIFICATION

Basic Standard: IEC 61000-4-11

Test duration time: Minimum three test events in sequence

Interval between event: Minimum 10 seconds

Angle: 0 / 45 / 90 / 135 / 180 / 225 / 270 / 315 degree

Step: 45 degree

8.9.2. TEST INSTRUMENT

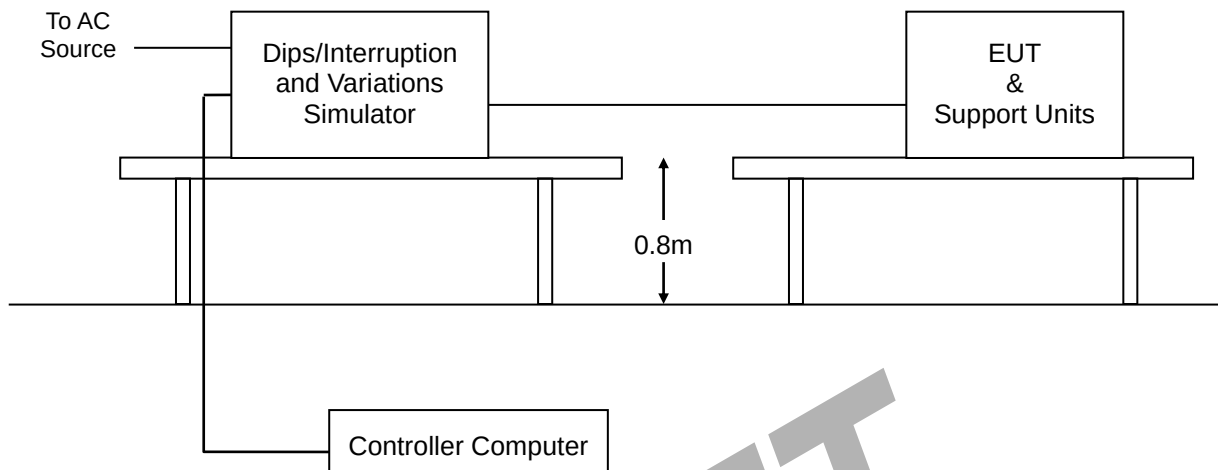
Immunity Shield Room					
Name of Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due
EMC Immunity Tester	EMC Partner	TRA2006	1144	2023/10/06	2024/10/05
CDN	EMC Partner	CDN-UTP8 ED3	046	2023/10/06	2024/10/05
Clamp	EMC Partner	CN-EFT1000	683	2024/08/02	2025/08/01
Software	N/A				

Note: The calibration interval of the above test instruments is 12 months and the calibrations are traceable to NML/ROC and NIST/USA.

8.9.3. TEST PROCEDURE (please refer to measurement standard or CCS SOP PA-028)

1. The EUT and support units were located on a wooden table, 0.8 m away from ground floor.
2. Setting the parameter of tests and then perform the test software of test simulator.
3. Conditions changes to occur at 0 degree crossover point of the voltage waveform.
4. Recording the test result in test record form.

8.9.4. TEST SETUP



- For the actual test configuration, please refer to the related item – Photographs of the Test Configuration.



Project No.: TM-2503000437P
Report No.: TMWK2503001385KE

Page: 67 / 77
Rev.: 00

8.9.5. TEST RESULTS

Temperature	23°C	Humidity	50% RH
Pressure	1001mbar	Tested By	Eason Liu
Tested Date	2024/09/03	Test Mode	Mode 1
Required Passing Performance	Criterion B: >95% reduction 0.5 periods Criterion C: 30% reduction 25 periods & >95% reduction 250 periods		

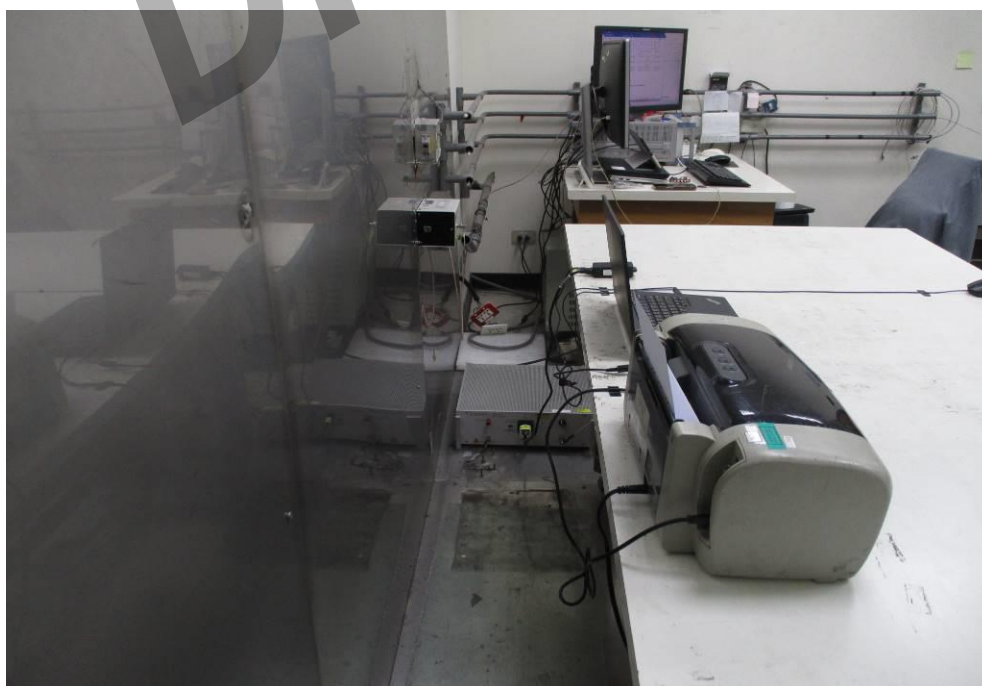
Test Power: 100/240Vac, 50Hz				
Voltage (% Reduction)	Duration (Period)	Performance Criterion	Observation	Test Result
>95	0.5	☒ A ☐ B ☐ C	Note ☒ 1 ☐ 2	PASS
30	25	☒ A ☐ B ☐ C	Note ☒ 1 ☐ 2	PASS
>95	250	☒ A ☐ B ☐ C	Note ☒ 1 ☐ 2	PASS

Note:
1. There was no change compared with initial operation during and after the test. No unintentional response was found during the test.

9 PHOTOGRAPHS OF THE TEST CONFIGURATION

CONDUCTED EMISSION TEST

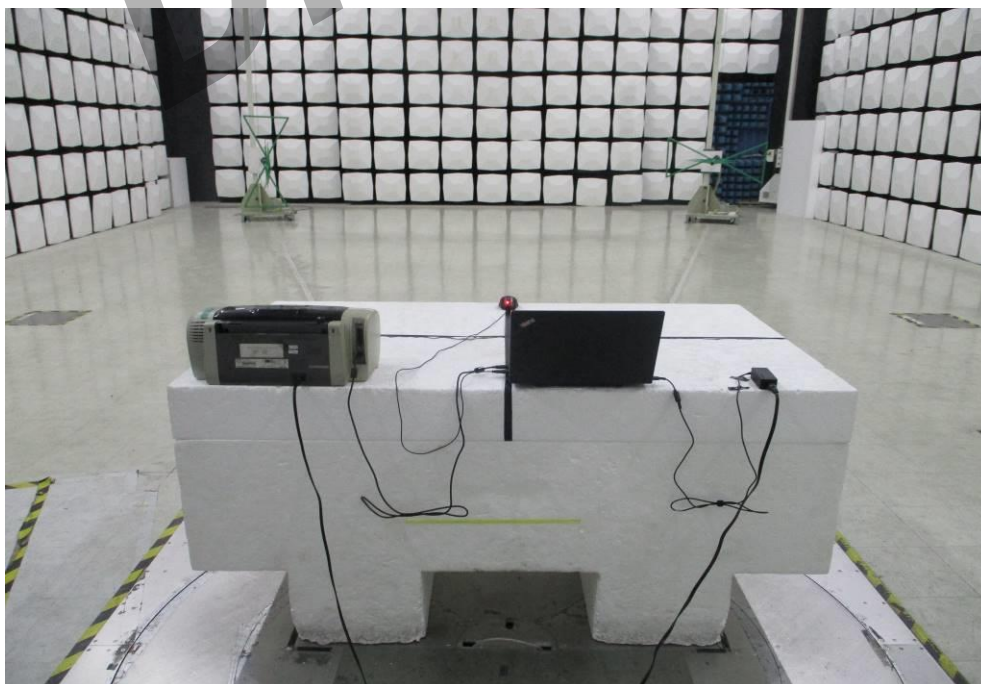
Mode 1



RADIATED EMISSION TEST

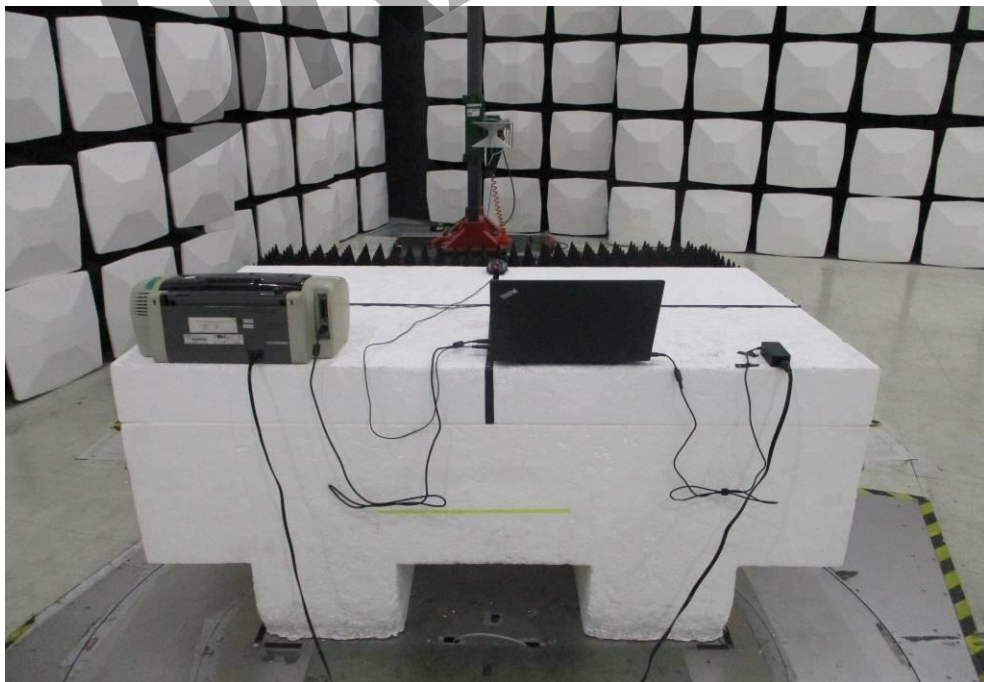
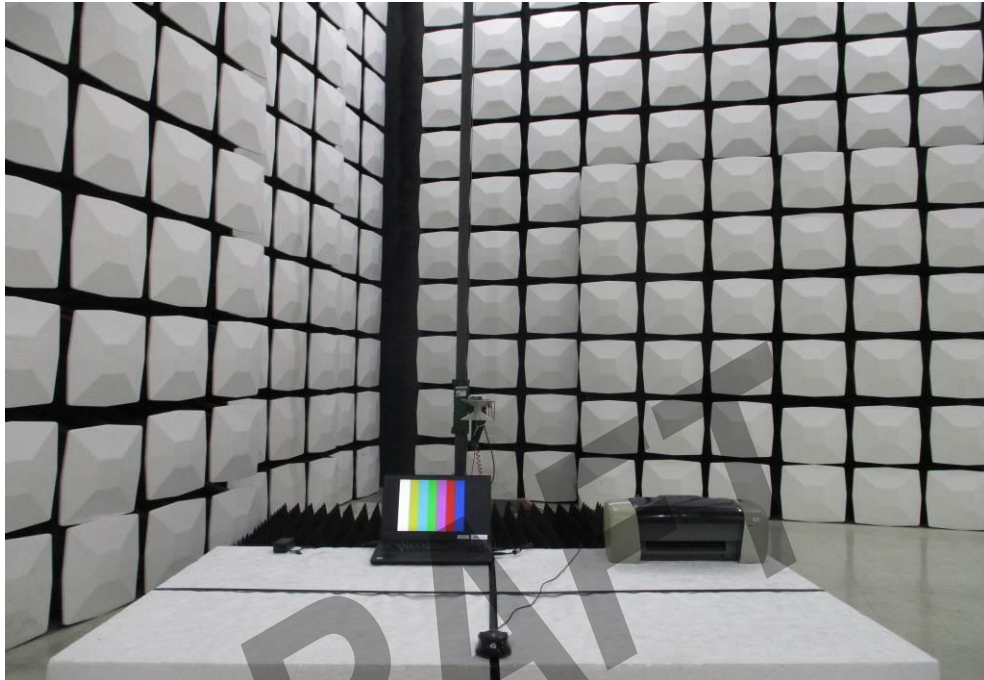
Below 1GHz

Mode 1



Above 1GHz

Mode 1



Harmonic & Flicker Test

Mode 1



ESD TEST

Mode 1



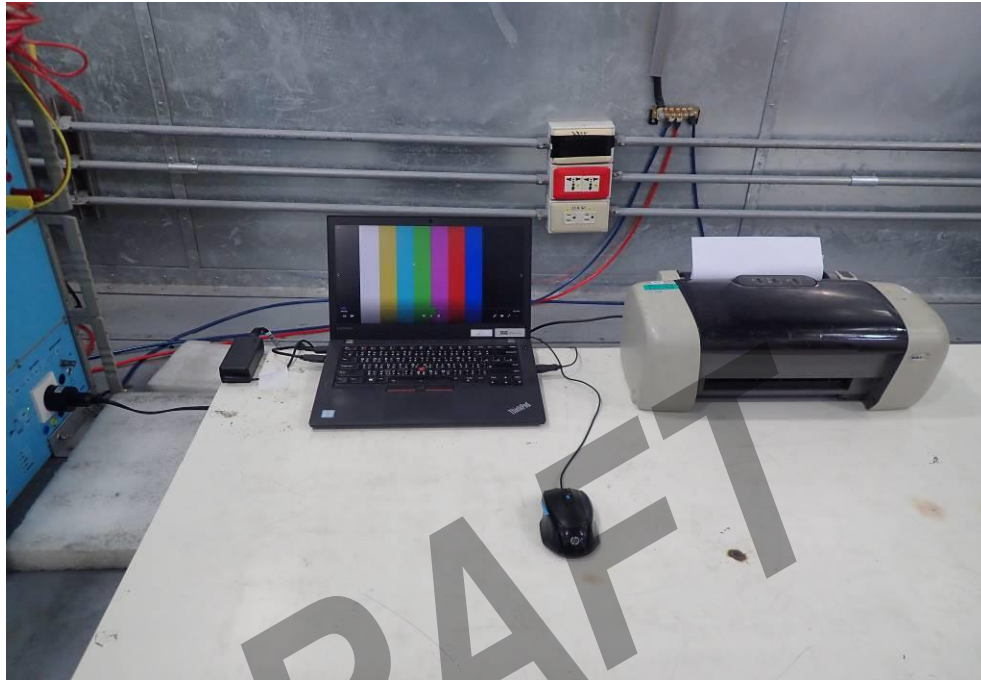
RS TEST

Mode 1



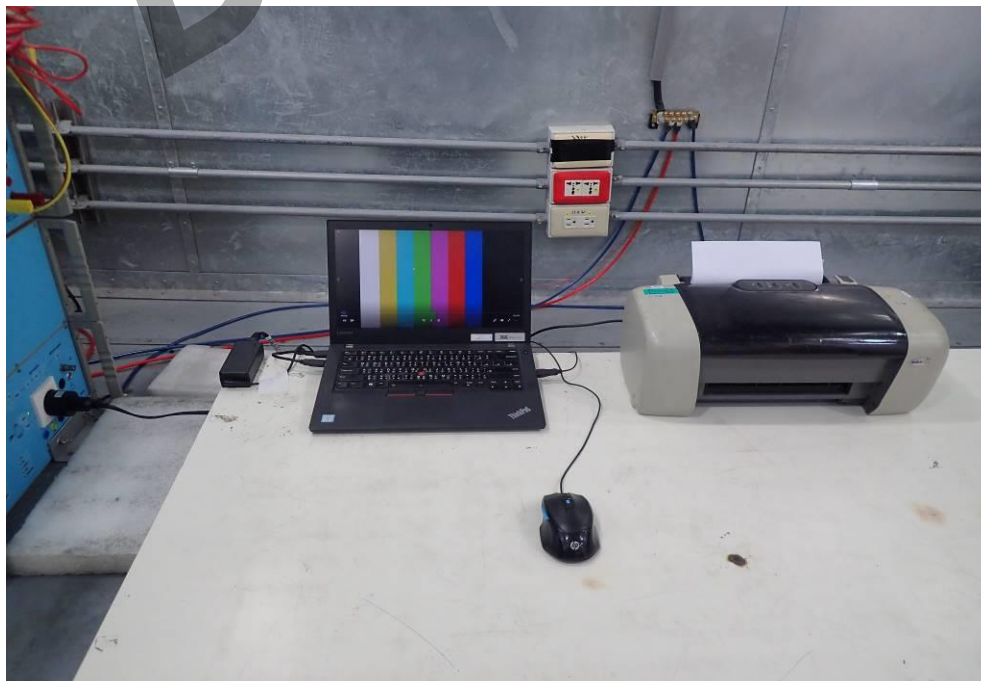
EFT Test

Mode 1



SURGE TEST

Mode 1



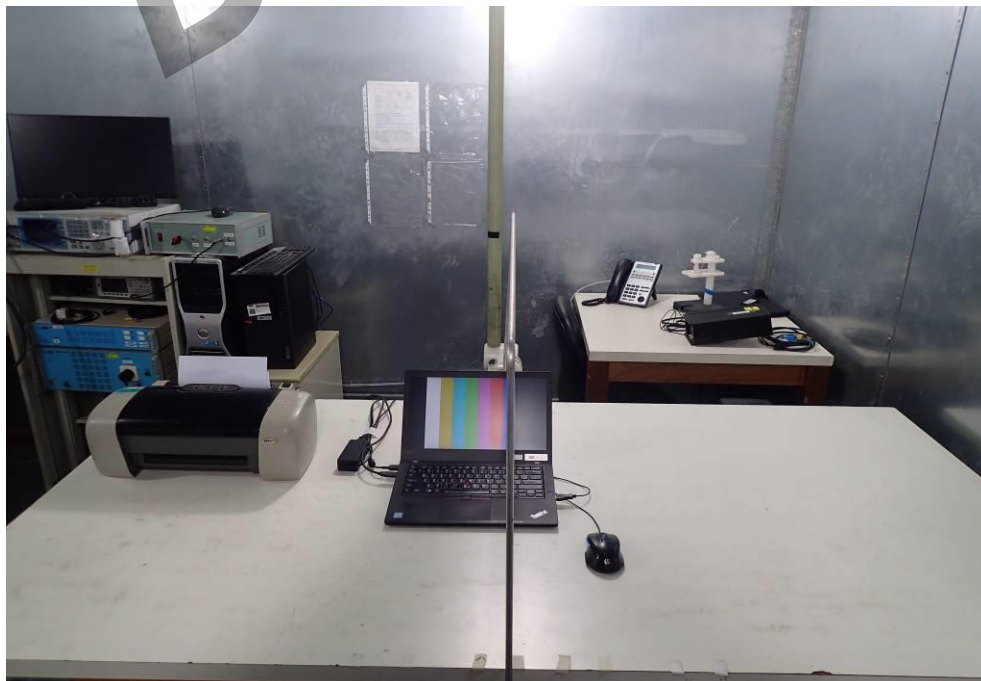
CS Test

Mode 1



POWER FREQUENCY MAGNETIC FIELD

Mode 1



VOLTAGE DIPS / INTERRUPTION TEST

Mode 1



APPENDIX 1 - PHOTOGRAPHS OF EUT

EXTERNAL PHOTOGRAPHS OF EUT

